

Lecture Notes of MATH97051

Manifolds

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0 Preliminaries

Definition 0.1. There are two points of view:

- **Extrinsic:** we think of our object inside an ambient space. ($\mathbb{R}^2$ or $\mathbb{R}^3$)
For example, circle of radius r , given by $\{(x, y) \mid x^2 + y^2 = r^2\}$.
- **Intrinsic:** we study our object independently of the ambient space.

Clarification 0.2. All spaces we study in this module will be Hausdorff and second-countable.

Remark 0.3. An atlas on S^2 via Stereographic Projection: $\{(S^2 \setminus \{S\}, f_1), (S^2 \setminus \{N\}, f_2)\}$ where

$$f_1 : S^2 \setminus \{S\} \rightarrow \mathbb{R}^2 : (x, y, z) \mapsto \left(\frac{x}{1+z}, \frac{y}{1+z} \right)$$

$$f_1^{-1} : \mathbb{R}^2 \rightarrow S^2 \setminus \{S\} : (X, Y) \mapsto \left(\frac{2X}{1+X^2+Y^2}, \frac{2Y}{1+X^2+Y^2}, \frac{1-X^2-Y^2}{1+X^2+Y^2} \right)$$

$$f_2 : S^2 \setminus \{N\} \rightarrow \mathbb{R}^2 : (x, y, z) \mapsto \left(\frac{x}{1-z}, \frac{y}{1-z} \right)$$

$$f_2^{-1} : \mathbb{R}^2 \rightarrow S^2 \setminus \{N\} : (X, Y) \mapsto \left(\frac{2X}{1+X^2+Y^2}, \frac{2Y}{1+X^2+Y^2}, \frac{-1+X^2+Y^2}{1+X^2+Y^2} \right)$$

and analogously we may derive an atlas for S^n via Stereographic Projection.

1 Topological Manifolds and Smooth Manifolds

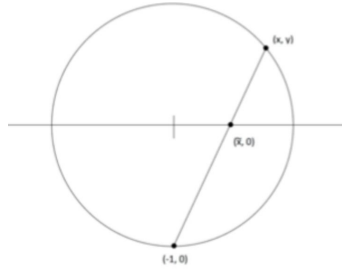
Definition 1.1. Let X be a topological space, a *co-ordinate chart* on X , denoted (U, f) is the data of:

- (1) An open set $U \subset X$
- (2) An open set $\tilde{U} \subset \mathbb{R}^n$ for some $n \geq 0$
- (3) A homeomorphism $f : U \rightarrow \tilde{U}$.

Definition 1.2. Let X be a topological space and assume that X is Hausdorff and second-countable. We say that X is a *topological manifold of dimension n* if for any point $x \in X$, there exists a co-ordinate chart (U, f) with $\tilde{U} = f(U) \subset \mathbb{R}^n$ and such that $x \in U$.

Remark 1.3. The value n here is not dependent on the point we choose.
(This is a result in algebraic topology and not required here.)

Example 1.4. (Circle) Consider the unit circle $S^1 = \{(x, y) \mid x^2 + y^2 = 1\} \subset \mathbb{R}^2$. We want to show that S^1 is a topological manifold. We first consider the open set $U_1 = S^1 \setminus \{(0, -1)\}$. Then the stereographic projection $f_1 : U_1 \rightarrow \mathbb{R}$ will give us a coordinate chart:



Fix $(x, y) \in S^1$, the line passing through (x, y) and $(0, -1)$ has equation $(X, Y) \in \mathbb{R}^2$ s.t. $\frac{Y+1}{y+1} - \frac{X}{x} = 0$. It intersects the X -axis at the point $\tilde{x} = \frac{x}{y+1}$. Thus, we define $f_1 : U_1 \rightarrow \mathbb{R}$ by

$$(x, y) \mapsto \tilde{x} = \frac{x}{y+1}$$

f_1 is invertible and the inverse is

$$f_1^{-1} : \mathbb{R} \rightarrow U_1 : \tilde{x} \mapsto \left(\frac{2\tilde{x}}{1+\tilde{x}^2}, \frac{1-\tilde{x}^2}{1+\tilde{x}^2} \right).$$

Both f_1 and f_1^{-1} are continuous, thus (U_1, f_1) defines a coordinate chart. Similarly, if $U_2 = S^1 \setminus \{(0, 1)\}$, then we show that $f_2 : U_2 \rightarrow \mathbb{R}$ such that (U_2, f_2) is a coordinate chart. Since $S^1 = U_1 \cup U_2$, it follows that S^1 is a topological manifold. Notice that the reason why we don't need to check that S^1 is Hausdorff and second-countable is because $S^1 \subset \mathbb{R}^3$.

Example 1.5. (Sphere) Consider the n -dimensional sphere $S^n = \{(x_0, \dots, x_n) \mid x_0^2 + \dots + x_n^2 = 1\} \subset \mathbb{R}^{n+1}$. Again, we consider the stereographic projection. Let $S = (0, \dots, 0, -1) \in S^n$ be the south pole, and $U_1 = S^n \setminus \{S\}$. Define the function $f_1 : U_1 \rightarrow \mathbb{R}^n$ by:

$$(x_0, \dots, x_n) \mapsto \left(\frac{x_1}{1+x_n}, \dots, \frac{x_n}{1+x_n} \right)$$

Then f_1 is continuous and invertible, with inverse

$$f_1^{-1} : \mathbb{R}^n \rightarrow U_1 : (\tilde{x}_0, \dots, \tilde{x}_{n-1}) \mapsto \left(\frac{2\tilde{x}_0}{1+\sum \tilde{x}_i^2}, \dots, \frac{2\tilde{x}_{n-1}}{1+\sum \tilde{x}_i^2}, \frac{1-\sum \tilde{x}_i^2}{1+\sum \tilde{x}_i^2} \right).$$

Thus (U, f_1) is a coordinate chart. Similarly, we can define $N = (0, \dots, 0, 1)$ the north pole and $U_2 = S^n \setminus \{N\}$. Then there exists $f_2 : U_2 \rightarrow \mathbb{R}^n$ such that (U_2, f_2) is a coordinate chart. Now since $S^n = U_1 \cup U_2$, S^n is a topological space (notice $S^n \subset \mathbb{R}^{n+1}$ is Hausdorff and second-countable).

Inspiration 1.6. Let X be a topological manifold and let $(U_1, f_1), (U_2, f_2)$ be two coordinate charts on X such that $U_1 \cap U_2 \neq \emptyset$. We have $f_1 : U_1 \rightarrow \tilde{U}_1 \subset \mathbb{R}^n$ and $f_2 : U_2 \rightarrow \tilde{U}_2 \subset \mathbb{R}^n$. Restricting them to the intersection, we get:

$$f_1 : U_1 \cap U_2 \rightarrow f_1(U_1 \cap U_2) \subset \tilde{U}_1 \subset \mathbb{R}^n \text{ and } f_2 : U_1 \cap U_2 \rightarrow f_2(U_1 \cap U_2) \subset \tilde{U}_2 \subset \mathbb{R}^n.$$

Definition 1.7. The *transition function* between these 2 coordinate charts is the function

$$\phi_{21} := f_2 \circ f_1^{-1} : f_1(U_1 \cap U_2) \xrightarrow{\sim} f_2(U_1 \cap U_2).$$

Example 1.8. (Circle) Consider the case of S^1 : $U_1 = S^1 \setminus \{(0, -1)\}$, $f_1^{-1} : \mathbb{R} \rightarrow U_1 : \tilde{x} \mapsto \left(\frac{2\tilde{x}}{1+\tilde{x}^2}, \frac{1-\tilde{x}^2}{1+\tilde{x}^2} \right)$. Also, $U_2 = S^1 \setminus \{(0, 1)\}$ and $f_2 : U_2 \rightarrow \mathbb{R} : (x, y) \mapsto \tilde{x} = \frac{x}{1-y}$. Thus, $U_1 \cap U_2 = S^1 \setminus \{(0, \pm 1)\}$ and $f_1(U_1 \cap U_2) = f_2(U_1 \cap U_2) = \mathbb{R} \setminus \{0\}$, $\phi_{21} = f_2 \circ f_1^{-1} : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R} \setminus \{0\} : \tilde{x} \mapsto 1/\tilde{x}$.

Remark 1.9. (Cocycle Properties) Let X be a topological manifold and let (U_i, f_i) be coordinate charts, with $i = 1, 2, 3$ such that $U_1 \cap U_2 \cap U_3 \neq \emptyset$. Then:

- (1) $\phi_{21} = f_2 \circ f_1^{-1} : f_1(U_1 \cap U_2) \rightarrow f_2(U_1 \cap U_2)$ is a homeomorphism
- (2) $\phi_{11} = \text{id}_{f_1(U_1)}$
- (3) $\phi_{12} = f_1 \circ f_2^{-1} = (f_2 \circ f_1^{-1})^{-1} = (\phi_{21})^{-1}$
- (4) $\phi_{31} = f_3 \circ f_1^{-1} = (f_3 \circ f_2^{-1}) \circ (f_2 \circ f_1^{-1}) = \phi_{32} \circ \phi_{21}$.

Notice in (4), we are considering all our f_i 's to be from $U_1 \cap U_2 \cap U_3 \rightarrow f_i(U_1 \cap U_2 \cap U_3)$.

Example 1.10. (Sphere) Let $S = (0, \dots, 0, -1) \in S^n$, $N = (0, \dots, 0, 1) \in S^n$. Let $U_1 = S^n \setminus \{S\}$ and $U_2 = S^n \setminus \{N\}$, we have $f_1(U_1 \cap U_2) = \mathbb{R}^n \setminus \{0\} = f_2(U_1 \cap U_2)$. We also had $f_2(x_0, \dots, x_n) = \left(\frac{x_1}{1-x_n}, \dots, \frac{x_{n-1}}{1-x_n} \right)$ and $f_1^{-1}(\tilde{x}_0, \dots, \tilde{x}_{n-1}) = \left(\frac{2\tilde{x}_0}{1+\sum \tilde{x}_i^2}, \dots, \frac{2\tilde{x}_{n-1}}{1+\sum \tilde{x}_i^2}, \frac{1-\sum \tilde{x}_i^2}{1+\sum \tilde{x}_i^2} \right)$. Therefore, the transition function ϕ_{21} is:

$$\phi_{21} = f_2 \circ f_1^{-1} : \mathbb{R}^n \setminus \{0\} \xrightarrow{\sim} \mathbb{R}^n \setminus \{0\} : (\tilde{x}_0, \dots, \tilde{x}_{n-1}) \mapsto \left(\frac{\tilde{x}_0}{\sum \tilde{x}_i^2}, \dots, \frac{\tilde{x}_{n-1}}{\sum \tilde{x}_i^2} \right).$$

Definition 1.11. Let X be a topological manifold. An *atlas* for X is a collection of coordinate charts $\{(U_i, f_i)\}_{i \in I}$ such that $\bigcup_{i \in I} U_i = X$. Note that the index I here might be infinite. If we are working with a compact manifold, then we can always choose a finite atlas; but assume the topological manifold to be non-compact, then we may choose a countable set I , this is guaranteed by X being second-countable.

Definition 1.12. A function $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is *smooth* (or lie in C^∞) if all the partial derivatives of F to any order exist.

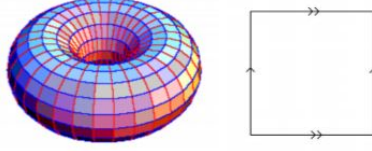
Definition 1.13. Let X be a topological manifold. Then an atlas $\{(U_i, f_i)\}$ on X is *smooth* if for any two coordinate charts in the atlas, the transition function $\phi_{ij} : f_j(U_i \cap U_j) \rightarrow f_i(U_i \cap U_j)$ is a smooth function for all $i, j \in I$. Note here both $f_j(U_i \cap U_j)$ and $f_i(U_i \cap U_j)$ are open. Also note that for i, j such that $U_i \cap U_j = \emptyset$, we automatically get ϕ_{ij} smooth.

Example 1.14. (Sphere) Recall $S = (0, \dots, 0, -1)$, $N = (0, \dots, 0, 1)$ in S^n , $U_1 = S^n \setminus \{S\}$, $U_2 = S^n \setminus \{N\}$. We have to show that the transition functions are smooth functions. (In general, it is not true that the inverse of a smooth function is smooth.) Now, $\phi_{21} = f_2 \circ f_1^{-1} : \mathbb{R}^n \setminus \{0\} \rightarrow \mathbb{R}^n \setminus \{0\} : (\tilde{x}_0, \dots, \tilde{x}_{n-1}) \mapsto \left(\frac{\tilde{x}_0}{\sum \tilde{x}_i^2}, \dots, \frac{\tilde{x}_{n-1}}{\sum \tilde{x}_i^2} \right)$. This function is clearly smooth since each coordinate is the ratio of two smooth functions, and the denominator is never zero. By symmetry, ϕ_{12} is also smooth, thus $\{(U_1, f_1), (U_2, f_2)\}$ is a smooth atlas.

Definition 1.15. Let $U, V \subset \mathbb{R}^n$ be open subsets and let $f : U \rightarrow V$ be a function. Then f is a *diffeomorphism* if f is smooth, it's a bijection, and the inverse f^{-1} is also smooth.

Remark 1.16. Recall that if ϕ_{ij} is a transition function then $\phi_{ij}^{-1} = \phi_{ji}$. Thus if $\{(U_i, f_i)\}_{i \in I}$ is a smooth atlas, then any transition function is a diffeomorphism.

Example 1.17. (Torus) Consider $\Lambda = \{(m, n) \mid m, n \in \mathbb{Z}\} \subset \mathbb{R}^2$. Let $\sim$ be the relation on $\mathbb{R}^2$ defined by $(x, y) \sim (x', y')$ if and only if $(x - x', y - y') \in \Lambda$ and let $T^2 = \mathbb{R}^2 / \sim = \{[(x, y)] \mid (x, y) \in \mathbb{R}^2\}$ be the quotient space with topology induced by $q : \mathbb{R}^2 \rightarrow T^2$. (the quotient map sends each point to its equivalence class and the topology on T^2 is defined such that a subset of T^2 is open iff its preimage in $\mathbb{R}^n$ is open, i.e. such that the quotient map is continuous.) Note that T^2 is Hausdorff and second-countable since $\mathbb{R}^2$ is so. Note also that $T^2 = [0, 1] \times [0, 1] / (x, 0) \sim (x, 1)$ and $(0, y) \sim (1, y)$.



We define $\tilde{U}_1 = (0, 1) \times (0, 1)$, $\tilde{U}_2 = (-\frac{1}{2}, \frac{1}{2}) \times (0, 1)$, $\tilde{U}_3 = (0, 1) \times (-\frac{1}{2}, \frac{1}{2})$ and $\tilde{U}_4 = (-\frac{1}{2}, \frac{1}{2}) \times (-\frac{1}{2}, \frac{1}{2})$. Also, let $U_i = q(\tilde{U}_i)$ for all i . Clearly U_i is open since its preimage is a union of open sets. Let

$$f_i = (q|_{\tilde{U}_i})^{-1} : U_i \rightarrow \tilde{U}_i \text{ for } i = 1, 2, 3, 4$$

Notice that $q|_{\tilde{U}_i}$ is clearly seen to be invertible. Verify that in two directions, open sets have open preimages, hence f_i is a homeomorphism, and consequently, (U_i, f_i) is a coordinate chart. Since $T^2 = \bigcup_{i=1}^4 U_i$, T^2 is a topological manifold. We now wish to show that $\{(U_i, f_i)\}_{i=1,2,3,4}$ is a smooth atlas. For ϕ_{21} :

$$f_1(U_1 \cap U_2) = \left(\left(0, \frac{1}{2}\right) \cup \left(\frac{1}{2}, 1\right) \right) \times (0, 1) \subset \tilde{U}_1$$

$$f_2(U_1 \cap U_2) = \left(\left(-\frac{1}{2}, 0\right) \cup \left(0, \frac{1}{2}\right) \right) \times (0, 1) \subset \tilde{U}_2$$

And the transition function is given by

$$\phi_{21} : f_1(U_1 \cap U_2) \rightarrow f_2(U_1 \cap U_2) : (x, y) \mapsto \begin{cases} (x, y) & \text{if } x < \frac{1}{2} \\ (x - 1, y) & \text{if } x > \frac{1}{2} \end{cases}$$

This is clearly a diffeomorphism. Checking that ϕ_{ij} is a diffeomorphism for any distinct $i, j \in \{1, 2, 3, 4\}$, it follows that $\{(U_i, f_i)\}_{i=1,2,3,4}$ is a smooth atlas.

Definition 1.18. Let X be a topological manifold and let $\mathcal{A} = \{(U_i, f_i)\}_{i \in I}$ be a smooth atlas on X . Let (U, f) be any co-ordinate chart on X . Then (U, f) is *compatible* with the atlas $\mathcal{A}$ if the transition function between (U, f) and any chart in $\mathcal{A}$ is a diffeomorphism. Thus, for any $i \in I$, the function $\phi : f \circ f_i^{-1} : f_i(U \cap U_i) \rightarrow f(U \cap U_i)$ is a diffeomorphism, given the non-empty condition.

Remark 1.19. (U, f) is compatible with a smooth atlas $\mathcal{A}$ iff $\mathcal{A} \cup \{(U, f)\}$ is a smooth atlas.

Proof. If (U, f) is compatible with $\mathcal{A}$, then for any transition function in $\mathcal{A} \cup \{(U, f)\}$, either it's a transition function in $\mathcal{A}$ or it's a transition function between (U, f) and a chart in $\mathcal{A}$. In the former case it's a diffeomorphism by definition of smooth atlas $\mathcal{A}$, and in the latter case it's again a diffeomorphism by assumption. Converse is trivial.

Definition 1.20. Let X be a topological manifold and let $\mathcal{A}$ and $\mathcal{B}$ be two smooth atlases. Then $\mathcal{A}$ and $\mathcal{B}$ are *compatible* if every chart in $\mathcal{B}$ is compatible with $\mathcal{A}$.

Remark 1.21. Smooth atlases $\mathcal{A}$ and $\mathcal{B}$ are compatible iff $\mathcal{A} \cup \mathcal{B}$ is a smooth atlas.

Proof. If $\mathcal{A}$ and $\mathcal{B}$ are compatible, take two charts in $\mathcal{A} \cup \mathcal{B}$, if they are both in $\mathcal{A}$, then the transition map is a diffeomorphism since $\mathcal{A}$ is a smooth atlas. Similarly if they are both in $\mathcal{B}$, the transition function is diffeomorphism. Now assume that one chart $U \in \mathcal{A}$ and the other $V \in \mathcal{B}$, since V is compatible with $\mathcal{A}$, the transition function between U, V is a diffeomorphism. Conversely, if $\mathcal{A} \cup \mathcal{B}$ is a smooth atlas, then $\mathcal{A}$ and $\mathcal{B}$ are clearly seen to be compatible.

Our goal is now to show that the notion of compatibility defines an equivalence relation.

Lemma 1.22. Let X be a topological manifold and let $\mathcal{A} = \{(U_i, f_i)\}_{i \in I}$ and $\mathcal{B} = \{(U_j, f_j)\}_{j \in J}$ be two compatible smooth atlases for X . Let (U, f) be a coordinate chart on X which is compatible with $\mathcal{A}$, then (U, f) is compatible with $\mathcal{B}$.

Proof. Fix $j \in J$, we need to show that the function $\phi = f_j \circ f^{-1} : f(U \cap U_j) \rightarrow f_j(U \cap U_j)$ is a diffeomorphism. Clearly ϕ is a homeomorphism since it's a composition of two homeomorphisms, we need to show that ϕ and ϕ^{-1} are smooth functions. Consider $\phi = f_j \circ f^{-1} : f(U \cap U_j) \rightarrow f_j(U \cap U_j)$. Let $y \in f(U \cap U_j)$, then

there exists a unique $x \in U \cap U_j$ such that $f(x) = y$. Since $\mathcal{A}$ is a smooth atlas on X , there exists $i \in I$ such that $x \in U_i$. So $x \in W := U \cap U_i \cap U_j$. We now have the homeomorphisms

$$f : W \rightarrow f(W), f_i : W \rightarrow f_i(W), f_j : W \rightarrow f_j(W)$$

In particular, $f_j \circ f^{-1} : f(W) \rightarrow f_j(W)$ is a homeomorphism such that $f_j \circ f^{-1} = (f_j \circ f_i^{-1}) \circ (f_i \circ f^{-1})$. Hence ϕ is smooth at $y = f(x)$ since it's the composition of two smooth functions $\implies \phi$ is a smooth function, and similarly ϕ^{-1} is smooth, so ϕ is a diffeomorphism.

Remark 1.23. We can now conclude that compatibility defines an equivalence relation on the set of smooth atlases on a given topological manifold.

Proof. Reflexivity and symmetry are obvious. If $\mathcal{A} \sim \mathcal{B}$ and $\mathcal{B} \sim \mathcal{C}$, then every chart in $\mathcal{C}$ is compatible with $\mathcal{B}$. Then by the previous lemma, it is also compatible with $\mathcal{A}$. Since every chart in $\mathcal{C}$ is compatible with $\mathcal{A}$, we see that $\mathcal{C}$ is compatible with $\mathcal{A} \implies \mathcal{A} \sim \mathcal{C}$.

Definition 1.24. A *smooth manifold* is a topological manifold X together with an equivalence class $[\mathcal{A}]$ of compatible smooth atlases on X . The equivalence class of atlases is called a *smooth structure* on X .

Remark 1.25. By lemma 1.22 it follows that if (U, f) is a chart compatible with $\mathcal{A}$, then (U, f) is compatible with any chart in $[\mathcal{A}]$. Consequently it makes sense to talk about coordinate charts compatible with structures or smooth manifolds.

Examples 1.26.

- Let $X = \mathbb{R}^n$ and $U = X$, $f = \text{id}_X$, then (U, f) is a coordinate chart and $\{(U, f)\}$ is a smooth atlas. Thus $\mathbb{R}^n$ is a smooth manifold. (unless otherwise specified, we put the smooth structure on $\mathbb{R}^n$ by default when talking about $\mathbb{R}^n$.)
- The sphere S^n is a smooth manifold, with the smooth structure found in example 1.14.
- The torus T^2 is a smooth manifold.

Remark 1.27. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a homeomorphism and let $U = X = \mathbb{R}^n$. Show that $\mathcal{A} = \{(U, f)\}$ is a smooth atlas on X which is compatible with the atlas defined in example 1.26 (1) iff f is a diffeomorphism.

Proof. $\mathcal{A}$ must be a smooth atlas since the only transition function is id_X . If $\mathcal{A}$ is compatible with $\{(U, \text{id}_X)\}$, then $f \circ \text{id}_X^{-1} = f$ is a diffeomorphism. Conversely if f is a diffeomorphism, then $\mathcal{A}$ is compatible with the trivial smooth atlas since the 2 transition functions f and f^{-1} are both diffeomorphisms.

Proposition 1.28. If X is a manifold and $V \subset X$ is an open subset of X , then V is also a manifold. If X is a smooth manifold with smooth atlas $\mathcal{A} = \{(U_i, f_i)\}_{i \in I}$ and $V \subset X$ is an open subset then V is also a smooth manifold.

Proof. (1) Consider an atlas $\{(U_i, f_i)\}_{i \in I}$ of X . For each $i \in I$, the restriction $f_i|_{U_i \cap V} : U_i \cap V \rightarrow f(U_i \cap V)$ is still a homeomorphism onto an open subset in $\mathbb{R}^n$. Since $\bigcup_i U_i = X$, we clearly get $\bigcup_i (U_i \cap V) = V$. Consequently, V is manifold. (2) Now we assume further that X is a smooth manifold with smooth structure $[\mathcal{A}]$. To prove that V is a smooth manifold, it suffices to find a smooth atlas for V . Consider $\mathcal{B} = \{(U_i \cap V, f_i|_{U_i \cap V})\}_{i \in I}$. The transition function

$$\phi_{ij} = f_i|_{U_i \cap V} \circ (f_j|_{U_j \cap V})^{-1} : f_j(U_i \cap U_j \cap V) \rightarrow f_i(U_i \cap U_j \cap V)$$

is clearly a diffeomorphism because it is a composition of the restrictions of two diffeomorphisms. Hence $\mathcal{B}$ is a smooth atlas and V a smooth manifold.

Notations 1.29. From now on, we omit the word “smooth” when it’s clear from the context, i.e. a manifold is a smooth manifold and an atlas is a smooth atlas. Moreover, a chart is a coordinate chart.

2 Submanifolds

Definition 2.1. Fix $n > 0$. An *affine subspace* $A \subset \mathbb{R}^n$ is a translation of a linear subspace of $\mathbb{R}^n$, i.e. there exists $v \in \mathbb{R}^n$ and a linear subspace $W \subset \mathbb{R}^n$ such that $A = \{v + w \mid w \in W\} \subset \mathbb{R}^n$.

Definition 2.2. Let X be an n -dimensional manifold and let $Y \subset X$ be a subset. Then Y is an *m -dimensional submanifold* of X if for all $y \in Y$, there exist

- (1) a co-ordinate chart (U, f) of X which is compatible with the smooth structure of X and s.t. $y \in U$;
- (2) an m -dimensional affine subspace $A \subset \mathbb{R}^n$ such that $f(U \cap Y) = f(U) \cap A$.

Proposition 2.3. Let X be an n -dimensional manifold and let Y be an m -dimensional submanifold. Then Y is an m -dimensional manifold.

Proof. STEP 1: We first show that Y is an m -dimensional topological manifold. Since X is a topological space which is Hausdorff and second countable, and Y inherits topology from X , Y is also Hausdorff and second countable. Now for all $y \in Y$, there is a coordinate chart (U, f) such that $y \in U$ and an affine subspace $A \subset \mathbb{R}^n$ such that $f(U \cap Y) = f(U) \cap A$. By linear algebra, there exists an invertible affine map $\tau : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that $\tau(A) = A_m$. Take $f' = \tau \circ f$ (note that f' is still compatible with the smooth structure on X), get $f'(U \cap Y) = f'(U) \cap A_m$, where $A_m = \{x_{m+1} = \dots = x_n = 0\} \subset \mathbb{R}^n$. Since $A_m \cong \mathbb{R}^m$, $(U \cap Y, f'|_{U \cap Y})$ defines a coordinate chart on Y , hence Y is an m -dimensional manifold.

STEP 2: We show that this induced atlas is indeed smooth. Let (U_1, f_1) and (U_2, f_2) be two charts compatible with the smooth structure of X with $f_i(U_i \cap Y) = f_i(U_i) \cap A_m$ for $i = 1, 2$. Let $V_i = U_i \cap Y$ and $g_i = f_i|_{V_i}$, then (V_i, g_i) is a chart on Y . By assumption the transition function

$$\phi_{21} = f_2 \circ f_1^{-1} : f_1(U_1 \cap U_2) \rightarrow f_2(U_1 \cap U_2)$$

is smooth and $f_i(U_i \cap Y) = f_i(U_i) \cap A_m$ for $i = 1, 2$. Since $g_i = f_i|_{V_i}$, we get the smoothness of

$$\psi_{21} = g_2 \circ g_1^{-1} : g_1(V_1 \cap V_2) \rightarrow g_2(V_1 \cap V_2).$$

Since all transition functions are smooth by such an argument, this atlas is smooth.

Theorem 2.4. (Inverse Function Theorem) Let $U \subset \mathbb{R}^n$ be an open subset and let $F : U \rightarrow \mathbb{R}^n$ be a smooth function. Let $x \in U$ be a point such that the Jacobian matrix at x $DF|_x : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is an isomorphism, then there exists an open neighbourhood $V \subset U$ of x such that the function $F|_V : V \rightarrow F(V)$ is a diffeomorphism.

Example 2.5. (Circle) Consider the circle $S^1 = \{(x, y) \mid x^2 + y^2 = 1\} \subset \mathbb{R}^2$. We showed that S^1 is a manifold in example 1.8, now we wish to show that it is also a submanifold of $\mathbb{R}^2$. Let:

$$U = \{(x, y) \mid x > 0\} \subset \mathbb{R}^2 \text{ and } f : U \rightarrow \mathbb{R}^2 : (x = r \cos \theta, y = r \sin \theta) \mapsto (r, \theta)$$

Then $\tilde{U} = f(U) = \{(r, \theta) \mid r > 0, \theta \in (-\frac{\pi}{2}, \frac{\pi}{2})\}$. Note that $f^{-1} : \tilde{U} \rightarrow U : (r, \theta) \mapsto (r \cos \theta, r \sin \theta)$ is smooth, and by inverse function theorem we get f is also smooth, so (U, f) is a coordinate chart on $\mathbb{R}^2$ such that

$$f(U \cap S^1) = \tilde{U} \cap \{(1, \theta) \mid \theta \in \mathbb{R}\} = \left\{ (1, \theta) \mid \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \right\}.$$

Similar claims hold for other standard open subsets of $\mathbb{R}^2$ ($x < 0$, $y > 0$, $y < 0$).

If we consider the affine map $\tau : (r, \theta) \mapsto (\theta, r - 1)$ and $f' = \tau \circ f$, then the restriction of f' to S^1 becomes $g = f'|_{U \cap S^1} : U \cap S^1 \rightarrow \mathbb{R}^2 : (\cos \theta, \sin \theta) \mapsto (\theta, 0)$.

Remark 2.6. $(U \cap S^1, g)$ in the above example defines a chart which is compatible with the smooth structure defined by the stereographic projection.

Proof. Denote $U_3 = U \cap S^1$ and $f_3 = g^1$, then $\phi_{13} = f_1 \circ (g^1)^{-1} : (-\frac{\pi}{2}, \frac{\pi}{2}) \rightarrow (0, \infty) : \theta \mapsto \frac{\cos \theta}{1 + \sin \theta}$ is a diffeomorphism. Similarly $\phi_{23} = f_2 \circ (g^1)^{-1} : (-\frac{\pi}{2}, \frac{\pi}{2}) \rightarrow (0, \infty) : \theta \mapsto \frac{\cos \theta}{1 - \sin \theta}$ is a diffeomorphism.

Definition 2.7. Let $h : \mathbb{R}^n \rightarrow \mathbb{R}^k$ be a smooth function and let $\alpha \in \mathbb{R}^k$. Then the set $Z_\alpha = \{x \in \mathbb{R}^n \mid h(x) = \alpha\} \subset \mathbb{R}^n$ is called the *level set* of h at α .

Examples 2.8.

- Let $n = 2$ and $k = 1$, $h = x_1^2 + x_2^2$. Then for any $\alpha > 0$, the level set of h at α is a circle, and in particular, a 1-diml submanifold of $\mathbb{R}^2$. If $\alpha = 0$ then the level set is a 0-diml submanifold.

- Let $n = 2$ and $k = 1$, $h = x_1x_2$. Then if $\alpha > 0$, the level set of h at α is 1-diml submanifold of $\mathbb{R}^2$ (consider the chart $(B_\varepsilon(x, y), f)$ where $f : \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}^2 \setminus \{0\} : (x, y) \mapsto (x, xy)$). But for $\alpha = 0$, it is not even a topological manifold, since there is no open neighbourhood of the origin that is homeomorphic to an open subset of $\mathbb{R}$.

Definition 2.9. Let $h : \mathbb{R}^n \rightarrow \mathbb{R}^k$ be a smooth function. A point $x \in \mathbb{R}^n$ is called a *regular point* of h if the Jacobian at x : $Dh|_x : \mathbb{R}^n \rightarrow \mathbb{R}^k$ is surjective. Notice that if $k > n$, then no point will be regular. $\alpha \in \mathbb{R}^k$ is a *regular value* if every point $x \in h^{-1}(\alpha)$ is a regular point. If $x \in \mathbb{R}^n$ is not a regular point, then it is called a *critical point*; if $\alpha \in \mathbb{R}^k$ is not a regular value, then it is called a *critical value*.

Example 2.10. Let $h(x_1, x_2) = x_1x_2$. Then $Dh|_{(x_1, x_2)} = (x_2, x_1)$ is not surjective at $(0, 0)$. Thus $\alpha = 0$ is a critical value.

Definition 2.11. Let $k \leq n$, the *standard projection* is the morphism $\pi : \mathbb{R}^n \rightarrow \mathbb{R}^k : (x_1, \dots, x_n) \mapsto (x_{n-k+1}, \dots, x_n)$, which forgets the first $n - k$ coordinates.

Theorem 2.12. (Implicit Function Theorem) Let $U \subset \mathbb{R}^n$ be an open subset and let $h : U \rightarrow \mathbb{R}^k$ be a smooth function, where $k \leq n$. Let $z \in U$ be a regular point of h , then there exists an open neighbourhood $V \subset U$ of z and a diffeomorphism $f : V \rightarrow f(V) \subset \mathbb{R}^n$ such that $h \circ f^{-1} = \pi : f(V) \rightarrow \mathbb{R}^k$ is the standard projection.

Proof. Let $x_1, \dots, x_n$ be coordinates on $\mathbb{R}^n$. By assumption, the Jacobian matrix $Dh|_z$ is a $k \times n$ matrix of rank k . Thus, after possibly changing the coordinates, we may assume that the last k columns form a basis of $\mathbb{R}^k$, i.e. the matrix

$$M := \begin{pmatrix} \frac{\partial h_1}{\partial x_{n-k+1}} \Big|_z & \cdots & \frac{\partial h_1}{\partial x_n} \Big|_z \\ \vdots & & \vdots \\ \frac{\partial h_k}{\partial x_{n-k+1}} \Big|_z & \cdots & \frac{\partial h_k}{\partial x_n} \Big|_z \end{pmatrix}$$

is invertible. Consider the function $f : U \rightarrow \mathbb{R}^n : x \mapsto (x_1, \dots, x_{n-k}, h_1(x), \dots, h_k(x))$. The Jacobian of f at z is the $n \times n$ matrix:

$$Df|_z = \left(\begin{array}{c|c} Id_{n-k} & 0 \\ \hline \star & M \end{array} \right).$$

Thus $Df|_z$ is invertible and by the inverse function theorem, there exists an open set $V \subset U$ containing z such that $f : V \rightarrow f(V) \subset \mathbb{R}^n$ is a diffeomorphism. If π is the standard projection, then $\pi \circ f = h \implies \pi = h \circ f^{-1}$.

Theorem 2.13. Let $h : \mathbb{R}^n \rightarrow \mathbb{R}^k$ be a smooth function and let $\alpha \in \mathbb{R}^k$ be a regular value of h . Then the level set $h^{-1}(\alpha) \subset \mathbb{R}^n$ is a submanifold of $\mathbb{R}^n$, of dimension $n - k$.

Proof. Let $z \in h^{-1}(\alpha)$. By definition, z is a regular point of h , then the implicit function theorem implies that there exists an open neighbourhood $V \subset U$ of z and a diffeomorphism $f : V \rightarrow f(V) \subset \mathbb{R}^n$ such that $h \circ f^{-1} = \pi : f(V) \subset \mathbb{R}^n \rightarrow \mathbb{R}^k$ is the standard projection. Then $f(h^{-1}(\alpha) \cap V) = f(V) \cap \pi^{-1}(\alpha)$. And since $\pi^{-1}(\alpha)$ is an affine subspace of $\mathbb{R}^n$ of dimension $n - k$, we are done.

Remark 2.14. Note that in this proof, we don't really need our smooth function h to be from $\mathbb{R}^n \rightarrow \mathbb{R}^k$. In fact, the theorem works for any smooth $h : U \rightarrow \mathbb{R}^k$ where U is an open subset of $\mathbb{R}^n$. In this case, the regular values are defined accordingly and the level set of a regular value under h is an $(n - k)$ -dimensional submanifold of $\mathbb{R}^n$.

Example 2.15. (Sphere) We want to show that the sphere S^n is a submanifold of $\mathbb{R}^{n+1}$. Let $h : \mathbb{R}^{n+1} \rightarrow \mathbb{R} : (x_0, \dots, x_n) \mapsto x_0^2 + \dots + x_n^2$. Then h is smooth and the Jacobian is $(2x_0, \dots, 2x_n)$. Thus any $\alpha \in \mathbb{R}_{>0}$ is a regular value. It follows that the sphere $x_0^2 + \dots + x_n^2 = \alpha$ is a submanifold of $\mathbb{R}^{n+1}$ of dimension n . We can verify that this is compatible with the smooth structure derived from stereographic projection:

$$f_1^{-1} : \mathbb{R}^n \rightarrow S^n \setminus \{S\} : (\tilde{x}_0, \dots, \tilde{x}_{n-1}) \mapsto \left(\frac{2\tilde{x}_0}{1 + \sum \tilde{x}_i^2}, \dots, \frac{2\tilde{x}_{n-1}}{1 + \sum \tilde{x}_i^2}, \frac{1 - \sum \tilde{x}_i^2}{1 + \sum \tilde{x}_i^2} \right)$$

$$f_3 : \{x_n > 0\} \subset \mathbb{R}^{n+1} \rightarrow f_3(\{x_n > 0\}) : (x_0, \dots, x_n) \mapsto (x_0, \dots, x_{n-1}, x_0^2 + \dots + x_n^2)$$

$$g_3 = f_3|_{h^{-1}(1)} : \{x_n > 0, \text{ on } S^n\} \rightarrow \{|x| < 1\} : (x_0, \dots, x_n) \mapsto (x_0, \dots, x_{n-1})$$

Consequently $\phi_{31} = g_3 \circ f_1^{-1} : \{|x| < 1\} \rightarrow \{|x| < 1\} : (\tilde{x}_0, \dots, \tilde{x}_{n-1}) \mapsto \left(\frac{2\tilde{x}_0}{1 + \sum \tilde{x}_i^2}, \dots, \frac{2\tilde{x}_{n-1}}{1 + \sum \tilde{x}_i^2} \right)$ is smooth.

Theorem 2.16. (Sard's) Let $h : \mathbb{R}^n \rightarrow \mathbb{R}^k$ be a smooth function, then the set of regular values $Z \subset \mathbb{R}^k$ is dense. More precisely, the set $\mathbb{R}^k \setminus Z$ has Lebesgue measure zero.

Notice here if $\alpha \notin h(\mathbb{R}^n)$, then α is a regular value. Thus, if $h : \mathbb{R}^n \rightarrow \mathbb{R}^k$, then “most of” the $\alpha \in \mathbb{R}^k$ are regular values, even if $k > n$.

Example 2.16. Let $h : \mathbb{R}^{n+1} \rightarrow \mathbb{R} : (x_0, \dots, x_n) \mapsto x_0^2 + \dots + x_n^2$ and let $\alpha < 0$. Then $h^{-1}(\alpha) = \emptyset$ is a submanifold of dimension n .

3 Smooth Functions

Definition 3.1 Let X be a manifold and let $h : X \rightarrow \mathbb{R}$ be a function. Then we say that h is *smooth* at $x \in X$ if for any chart (U, f) , which is compatible with the smooth structure of X and such that $x \in U$, the function $h \circ f^{-1} : f(U) \rightarrow \mathbb{R}$ is smooth at the point $f(x)$. We say that h is *smooth* if h is smooth at every point x .

Lemma 3.2. Let X be a manifold and let $h : X \rightarrow \mathbb{R}$ be a function. Let (U_1, f_1) and (U_2, f_2) be coordinate charts which are compatible with the smooth structure of X . Let $x \in U_1 \cap U_2$, then $h \circ f_1^{-1}$ is smooth at $f_1(x)$ if and only if $h \circ f_2^{-1}$ is smooth at $f_2(x)$.

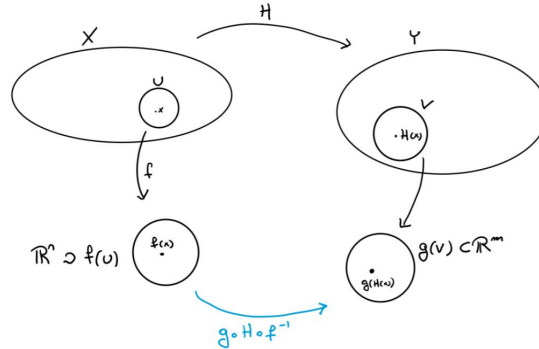
Proof. We assume that $h \circ f_2^{-1}$ is smooth at $f_2(x)$. By assumption, the transition function $\phi_{21} : f_2 \circ f_1^{-1} : f_1(U_1 \cap U_2 \cap U) \rightarrow f_2(U_1 \cap U_2 \cap U)$ is a diffeomorphism such that $\phi_{21}(f_1(x)) = f_2(x)$ (here (U, f) is some chart in the atlas of X with $x \in U$). Thus, $h \circ f_1^{-1} = (h \circ f_2^{-1}) \circ (f_2 \circ f_1^{-1}) = (h \circ f_2^{-1}) \circ \phi_{21}$ is smooth at $f_1(x)$. The converse is completely symmetric.

Remark. Now the definition of smooth function does not depend on the choice of the coordinate chart.

Definition 3.3. Let X be a manifold of dimension n and Y be a manifold of dimension m . Then a function $H : X \rightarrow Y$ is *smooth* at x if there exist

- (1) a chart (U, f) compatible with the smooth structure of X , where $x \in U$ and
- (2) a chart (V, g) compatible with the smooth structure of Y , where $H(x) \in V$

such that $H(U) \subset V$ and the function $g \circ H \circ f^{-1} : f(U) \rightarrow g(V)$ is smooth at $f(x)$. We say that H is *smooth* if it is smooth at all points of X .



Remark. The condition $H(U) \subset V$ is not strong since in most cases we may have a continuous H . Restricting f to the intersection of U and the preimage of V under H (still an open set in U) will work.

Remark. This definition does not depend on the charts on X and Y : let $(U_1, f_1), (U_2, f_2)$ be two charts on X and $x \in U_1 \cap U_2$; let $(V_1, g_1), (V_2, g_2)$ be two charts on Y such that $H(x) \in V_1 \cap V_2$. Require $H(U_1) \subset V_1$ and $H(U_2) \subset V_2$. If $g_1 \circ H \circ f_1^{-1}$ is smooth at $f_1(x)$, then $g_2 \circ H \circ f_2^{-1} = \psi_{21} \circ g_1 \circ H \circ f_1^{-1} \circ \phi_{12}$ (on $f_2(U_1 \cap U_2)$) is smooth at the point $f_2(x)$, where ψ_{21} and ϕ_{12} are relevant transition functions. (If the charts are compatible with instead of lying in the smooth atlases we intersect $U_1 \cap U_2$ with U , $V_1 \cap V_2$ with V .)

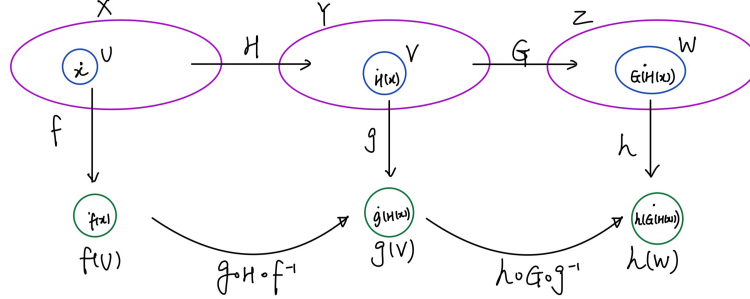
Definition 3.4. We say that H is a *diffeomorphism* if H is smooth, it's a bijection and its inverse is smooth. In this case the manifolds X and Y are said to be *diffeomorphic*.

Lemma 3.5. Let $Y \subset X$ be a submanifold and let $i : Y \hookrightarrow X$ be the inclusion morphism, i.e. $i(y) = y$ for all $y \in Y$. Then i is smooth.

Proof. Let $y \in Y$. Then by the definition of submanifold, there exists a chart (V, g) with $y \in V \subset X$ such

that $g(V \cap Y) = g(V) \cap A$ for some affine subspace $A \subset \mathbb{R}^n$. Let $U = V \cap Y$ and $f = g|_U$. Then (U, f) is a chart on Y such that $y \in U$ (this is how we construct the smooth atlas on Y so automatically smooth). We have $g \circ i \circ f^{-1} = \text{id}_{f(U)}$. Thus i is smooth at y .

Lemma 3.6. Let X, Y, Z be manifolds and $H : X \rightarrow Y, G : Y \rightarrow Z$ be smooth functions. Then $G \circ H$ is also a smooth function.



Proof. Fix $x \in X$. By assumption, there exists a chart (U, f) on X such that $x \in U$ and a chart (V, g) on Y such that $H(x) \in V$, and $g \circ H \circ f^{-1}$ is smooth at $f(x)$. Similarly, there exists a chart (W, h) on Z such that $G(H(x)) \in W$ and such that $h \circ G \circ g^{-1}$ is smooth at $g(H(x))$. Note that we can take the chart on Y to be (V, g) wlog, since we can always restrict to the intersection and composition with transition function is still smooth. Note also that $g(H(x)) = (g \circ H \circ f^{-1})(f(x))$. Therefore, $h \circ G \circ H \circ f^{-1} = (h \circ G \circ g^{-1}) \circ (g \circ H \circ f^{-1})$ (on a nbhd of $f(x)$) is smooth at $f(x)$, since it is the composition of two smooth functions.

Lemma 3.7. Let $H : X \rightarrow Y$ be a smooth function between manifolds and let $Z \subset Y$ be a submanifold. Assume that the image of H is contained in Z . Then H defines a smooth function $X \rightarrow Z$.

Proof. Fix $x \in X$. Then there exist a chart (U, f) on X and a chart (V, g) on Y such that $x \in U$ such that $H(U) \subset V$ and $g \circ H \circ f^{-1}$ is smooth. Now let $W = V \cap Z$ and $\tilde{g} = g|_W$. Clearly $(W, \tilde{g})$ is a chart on Z and $\tilde{g} \circ H \circ f^{-1}$ is equal to $g \circ H \circ f^{-1}$ on X , so it's smooth at $f(x)$.

Lemma 3.8. If V is a vector space of dimension n over $\mathbb{R}$, then V admits a smooth structure such that V is diffeomorphic to $\mathbb{R}^n$.

Proof. Let ϕ be the correspondence map from $V \rightarrow \mathbb{R}^n$. Clearly $\text{id}_{\mathbb{R}^n} \circ \phi \circ \phi^{-1} = \text{id}_{\mathbb{R}^n}$ is smooth. Hence $\phi : V \rightarrow \mathbb{R}^n$ is smooth. Proving smoothness of ϕ^{-1} is completely analogous.

Remark. If we change a choice of basis, say (V, ϕ_1) and (V, ϕ_2) , these 2 charts are compatible since $\phi_1 \circ \phi_2^{-1}$ gives a linear isomorphism, which is clearly a diffeomorphism.

4 Product of Manifolds

Construction 4.1. Let X and Y be two manifolds of dimension n and m respectively. Then the product $X \times Y$ is a manifold of dimension $m+n$. Indeed, let $(x, y) \in X \times Y$, then there exists a chart (U, f) on X and a chart (V, g) on Y such that $x \in U, y \in V$ and $f(U) \subset \mathbb{R}^n, g(V) \subset \mathbb{R}^m$ are open sets. Thus if $W = U \times V$ and $h = (f, g) : W \rightarrow h(W) = f(U) \times g(V) \subset \mathbb{R}^{m+n}$, then (W, h) is a chart such that $(x, y) \in W$.

Remark. To show that $X \times Y$ is a smooth manifold, we need to prove smoothness of the transition function. The transition function $(f_1, g_1) \circ (f_2, g_2)^{-1} : f_2(U_1 \cap U_2) \times g_2(U_1 \cap U_2) \rightarrow f_1(U_1 \cap U_2) \times g_1(U_1 \cap U_2)$ takes (x, y) to $(f_1 f_2^{-1}(x), g_1 g_2^{-1}(y))$. Notice that each coordinate function is smooth since X, Y are smooth manifolds, hence the transition function is smooth. Its inverse is smooth by a symmetric argument so the transition function is a diffeomorphism.

Definition 4.2. A *Lie Group* is a manifold G which has a group structure $(G, \cdot)$ such that the multiplication function $m : G \times G \rightarrow G : (g, h) \mapsto g \cdot h$ and the inverse function $i : G \rightarrow G : g \mapsto g^{-1}$ are both smooth functions (of manifolds).

General Linear Group.

We denote by $M(n, \mathbb{R})$ the set of $n \times n$ matrices with real coefficients. Then $M(n, \mathbb{R})$ is a vector space of dimension n^2 over $\mathbb{R}$. Consequently it is an n^2 -dimensional manifold. Let

$$GL(n, \mathbb{R}) := \{A \in M(n, \mathbb{R}) \mid \det(A) \neq 0\}.$$

Then $GL(n, \mathbb{R})$ is an n^2 -dimensional manifold, since it is an open subset of $M(n, \mathbb{R})$ (due to continuity of the determinant function). $GL(n, \mathbb{R})$ is also a group under multiplication, we want to check that the product and the inverse morphisms are smooth:

- Let $A, B \in GL(n, \mathbb{R})$, then each coefficient in $A \cdot B$ is a polynomial in the coefficients of A and B . Thus it defines a smooth function.
- Similarly, if $A \in GL(n, \mathbb{R})$, then each coefficient of A^{-1} is a rational function in coefficients of A , this is also a smooth function since $\det(A) \neq 0$.

5 More on Smooth Functions

Definition 5.1. Let X be a manifold of dimension n and let Y be a manifold of dimension k . Let $H : X \rightarrow Y$ be a smooth function. Take $x \in X$ and a chart (U, f) on X at x , and let (V, g) be a chart at $H(x)$ such that $H(U) \subset V$. Now, by definition of smoothness of H , the function $\tilde{H} := g \circ H \circ f^{-1} : f(U) \rightarrow g(V)$ is smooth at $f(x)$ and its Jacobian defines a linear map $D\tilde{H}|_{f(x)} : \mathbb{R}^n \rightarrow \mathbb{R}^k$. Define the *rank of H at x* to be the rank of the matrix $D\tilde{H}|_{f(x)}$.

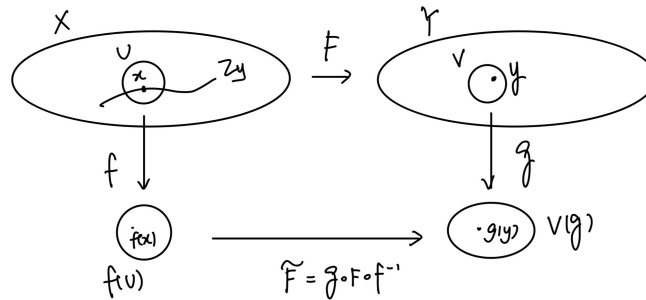
Remark. rank is well defined. In other words, it does not depend on the choice of charts on X and Y . Consider a new chart (U', f') at x and (V', g') at $H(x)$ such that $H(U') \subset V'$. Then the function $\tilde{H}' = g' \circ H \circ (f')^{-1} : f'(U') \rightarrow g'(V')$ is smooth at $f'(x)$ (proven before, smoothness of a function at a point is well-defined). Now let $\phi = f \circ (f')^{-1} : f'(U \cap U') \rightarrow f(U \cap U')$ and $\psi = g \circ (g')^{-1} : g'(V \cap V') \rightarrow g(V \cap V')$ be the transition functions. (again, wlog we assume that the charts are in atlas instead of just compatible)

$$\tilde{H} = g \circ H \circ f^{-1} \text{ and } \tilde{H}' = g' \circ H \circ (f')^{-1}$$

Thus, $\tilde{H}' = \psi^{-1} \circ \tilde{H} \circ \phi$ on $f'(U \cap U') \rightarrow g'(V \cap V')$ and note that $\phi(f'(x)) = f(x)$ and $\tilde{H}(f(x)) = g(H(x))$. Therefore, by the chain rule, $D\tilde{H}'|_{f'(x)} = D\psi^{-1}|_{g(H(x))} \circ D\tilde{H}|_{f(x)} \circ D\phi|_{f'(x)}$. Since $D\psi^{-1}|_{g(H(x))}$ and $D\phi|_{f'(x)}$ are invertible, the rank of $D\tilde{H}|_{f(x)}$ is equal to that of $D\tilde{H}'|_{f'(x)}$.

Definition 5.2. Let X be a manifold of dimension n and Y be a manifold of dimension k . Let $F : X \rightarrow Y$ be a smooth function. Then $x \in X$ is a *regular point* if the rank of F at x is equal to k . (In other words, the Jacobian matrix of F at x is surjective. Note that for this to hold we must have $n \geq k$.) If x is not regular, then it is called a *critical point*. A point $y \in Y$ is called a *regular value* if every point $x \in F^{-1}(y)$ is a regular point, otherwise it is called a *critical value*. Note that if the preimage of a value is empty, then we automatically get a regular value since there's no point to check.

Lemma 5.3. Let X be a manifold of dimension n and Y be a manifold of dimension k . Let $F : X \rightarrow Y$ be a smooth function and $y \in Y$ a regular value of F . Then the level set $Z_y = F^{-1}(y) \subset X$ is a submanifold of X of dimension $n - k$.



Proof. Let $x \in Z_y$, then there exists a chart (U, f) with $x \in U \subset X$ and a chart (V, g) with $F(x) \in V \subset Y$ and $F(U) \subset V$. We have the function $\tilde{F} = g \circ F \circ f^{-1} : f(U) \rightarrow g(V)$ and $\tilde{F}^{-1}(g(y)) = f(Z_y \cap U)$. Since $y \in Y$ is a regular value of F , $f(x)$ is a regular point of $\tilde{F}$. Since x is arbitrary in $Z_y \cap U$, we have $g(y)$ a regular value of $\tilde{F}$. Therefore, the level set $\tilde{F}^{-1}(g(y))$ is a submanifold of $\mathbb{R}^n$. This means that there exists a chart (W, h) with $f(x) \in W \subset f(U)$ such that $h(\tilde{F}^{-1}(g(y)) \cap W) = h(W) \cap A$. It follows that $(f^{-1}(W), h \circ f)$ is a coordinate chart on X such that $h \circ f(f^{-1}(W) \cap Z_y) = h \circ f(f^{-1}(W)) \cap A$. Thus Z_y is a submanifold of X .

Example 5.4. Let $S^n = \{(x_0, \dots, x_n) \mid \sum x_i^2 = 1\} \subset \mathbb{R}^{n+1}$ and consider the function $F : S^n \rightarrow \mathbb{R}$ sending $(x_0, \dots, x_n) \mapsto x_0$. Now since F is the composition of $i : S^n \rightarrow \mathbb{R}^{n+1}$ and $\pi_0 : \mathbb{R}^{n+1} \rightarrow \mathbb{R}$, it is a smooth function. We want to find the regular points of F . Consider the chart given by $U_n = S^n \cap \{x_n > 0\}$, and the function $f_n : U_n \rightarrow \mathbb{R}^n : (x_0, \dots, x_n) \mapsto (x_0, \dots, x_{n-1})$. We have $f_n(U_n) = \{(x_0, \dots, x_{n-1}) \mid \sum_{i=0}^{n-1} x_i^2 < 1\}$. Then $\tilde{F}_n = F \circ f_n^{-1} : f_n(U_n) \rightarrow \mathbb{R} : (x_0, \dots, x_{n-1}) \mapsto x_0$ and $D\tilde{F}_n = (1, 0, \dots, 0)$. Thus every point in U_n is a regular point. Now consider $U_0 = S^n \cap \{x_0 > 0\}$ and $f_0 : U_0 \rightarrow \mathbb{R}^n : (x_0, \dots, x_n) \mapsto (x_1, \dots, x_n)$. We have $f_0(U_0) = \{(x_1, \dots, x_n) \mid \sum_{i=1}^n x_i^2 < 1\}$. $\tilde{F}_0 = F \circ f_0^{-1} : f_0(U_0) \rightarrow \mathbb{R} : (x_1, \dots, x_n) \mapsto +\sqrt{1 - \sum_{i=1}^n x_i^2}$.

$$D\tilde{F}_0|_{(x_1, \dots, x_n)} = -\frac{(x_1, \dots, x_n)}{\sqrt{1 - \sum_{i=1}^n x_i^2}}$$

Thus the point $(1, 0, \dots, 0) \in S^n$ is a critical point. By symmetry, it follows that the only two critical values are ± 1 . To sum up: $F^{-1}(\alpha) = (n-1)$ -dimensional sphere if $\alpha \in (-1, 1)$, a point if $\alpha = \pm 1$ and $\emptyset$ otherwise.

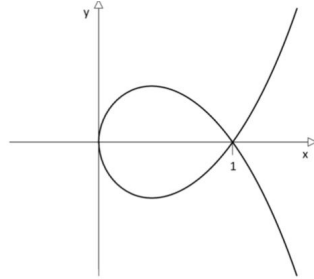
Definition 5.5. Let X be a manifold of dimension n and Y be a manifold of dimension k . A smooth function $F : X \rightarrow Y$ is said to be a *submersion* if the rank of F at any point $x \in X$ is equal to k (in this case, must have $k \leq n$); F is an *immersion* if the rank of F at any point is equal to n (in this case, $n \leq k$).

Remark. F is an immersion at $x \in X$ iff the Jacobian at x is injective; F is a submersion at $x \in X$ iff the Jacobian at x is surjective iff x is a regular point of F .

Lemma 5.6. Let $Y \subset X$ be an m -dimensional submanifold. Then the inclusion $i : Y \hookrightarrow X$ is an immersion.

Proof. This is very similar to lemma 3.5 (where we showed such i is smooth). Let $y \in Y$, then there exists a chart (V, g) with $y \in V \subset X$ such that $g(V \cap Y) = g(V) \cap A$ for some affine subspace $A \subset \mathbb{R}^n$. Let $U = V \cap Y$ and $f = g|_U$, then (U, f) is a coordinate chart on Y such that $y \in U$ and $f(U) \subset A \cong \mathbb{R}^m$. We have $g \circ i \circ f^{-1} = \text{id}_{f(U)}$, which is smooth. By definition of a smooth function, the rank of i at y is the rank of $\text{id}_{f(U)}$ at $f(y)$, which is equal to $m = \dim Y$. Thus i is an immersion.

Example 5.7. Let $F : \mathbb{R} \rightarrow \mathbb{R}^2 : t \mapsto (t^2, t^3 - t)$. The Jacobian of F at $t \in \mathbb{R}$ is $DF|_t = (2t, 3t^2 - 1)$. It follows that $DF|_t \neq 0$ for all t , i.e. the rank of F is 1 for all $t \in \mathbb{R}$. Thus, F is an immersion. However, $F(\mathbb{R})$ is not a submanifold. (It is not a topological manifold since at $(1, 0)$, we can't find a chart).



Remark. A submanifold induces an immersion; but an immersion does not necessarily give a submanifold.

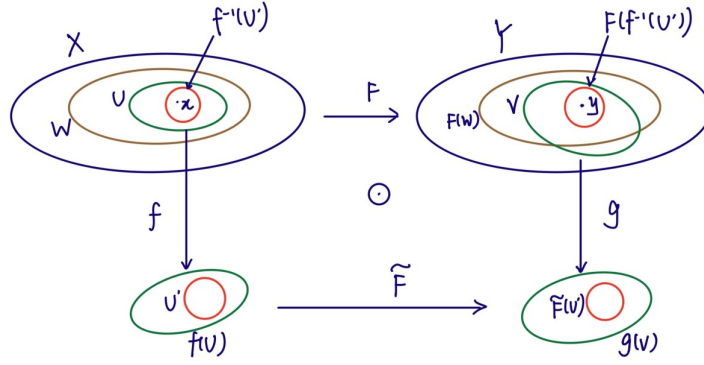
Proposition 5.8. Let X and Y be n -dimensional manifolds and let $F : X \rightarrow Y$ be a function which is smooth, bijective and of rank n at any $x \in X$. Then F is a diffeomorphism.

Proof. Fix $y \in Y$ and let $x \in F^{-1}(y)$. Let (U, f) be a chart on X with $x \in U$ and (V, g) be a chart on Y such that $f(U) \subset V$. Let $\tilde{F} = g \circ F \circ f^{-1}$. Then the Jacobian $D\tilde{F}|_{f(x)}$ has rank n . It follows that the Jacobian $D\tilde{F}^{-1}|_{g(y)}$ also has rank n . By inverse function theorem, it follows that $\tilde{F}^{-1}$ is smooth at $g(y)$, thus F^{-1} is smooth at y .

Lemma 5.9. Let X be a manifold of dimension n and Y be a manifold of dimension m . Let $p : X \times Y \rightarrow X : (x, y) \mapsto x$. Then p is a smooth function which is a submersion, i.e. the rank of p at any point is n .

Proof. Let us take a generic point $(x, y) \in X \times Y$. We get a chart (U, f) around x in X and a chart (V, g) around y in Y . Then we get a chart $(U \times V, (f, g))$ around (x, y) . Clearly $p(U \times V) = U$. Now let $\tilde{F} = g \circ p \circ (f, g)^{-1} : f(U) \times g(V) \rightarrow g(V) : (a, b) \mapsto b$. The Jacobian matrix $D\tilde{F}|_{(x, y)} = (0_n | I_n)$ clearly has rank n , so p is a submersion.

Proposition 5.10. Let X be a manifold of dimension n and Y be a manifold of dimension k . Let $F : X \rightarrow Y$ be a submersion, then $F : X \rightarrow Y$ is an open map, i.e. for any open subset $W \subset X$, $F(W)$ is open in Y .



Proof. Let $W \subset X$ be an open subset and let $y \in F(W)$. Take some $x \in F^{-1}(y) \cap W$. Let (U, f) be a chart at x and let (V, g) be a chart at y such that $F(U) \subset V$. We may assume that $U \subset W$. Since F is a submersion, the function $\tilde{F} = g \circ F \circ f^{-1} : f(U) \rightarrow g(V)$ is smooth of rank k , i.e. $f(x)$ is a regular point for $\tilde{F}$. By the implicit function theorem, there exists an open set $U' \subset f(U)$ containing $f(x)$ and a diffeomorphism $h : U' \rightarrow h(U') \subset \mathbb{R}^n$ such that $\tilde{F} \circ h^{-1} : h(U') \rightarrow \mathbb{R}^k$ is the standard projection. In particular, $\tilde{F}(U') = \tilde{F} \circ h^{-1} \circ h(U')$ is open in $g(V)$. Since g and f are both homeomorphisms, it follows that $F(f^{-1}(U')) \subset V$ is open, thus $F(f^{-1}(U')) \subset Y$ is open. Moreover, since $U' \subset f(U)$, we get $f^{-1}(U') \subset U \subset W$. So $y \in F(f^{-1}(U')) \subset F(W)$. $F(W)$ contains an open nbhd of y . Consequently $F(W) \subset Y$ is open.

6 Tangent Spaces

Need an intrinsic point of view: we want a definition of a tangent vector that doesn't depend on the embedding of a manifold in a given ambient space.

Definition 6.1. Let U be open in $\mathbb{R}^n$ and $x \in U$. A *curve* through x is a smooth function $\sigma : (-\varepsilon, \varepsilon) \rightarrow U$ where $\varepsilon > 0$ and $\sigma(0) = x$. The Jacobian $D\sigma|_0$ of σ at 0 is the *tangent vector* of σ at point x .

Definition 6.2. Let $\sigma_1 : (-\varepsilon_1, \varepsilon_1) \rightarrow U$ and $\sigma_2 : (-\varepsilon_2, \varepsilon_2) \rightarrow U$ be two curves in $\mathbb{R}^n$ passing through x , i.e. $\sigma_1(0) = \sigma_2(0) = x$. We write $\sigma_1 \sim \sigma_2$ if and only if $D\sigma_1|_0 = D\sigma_2|_0$, i.e. $\sigma_1 \sim \sigma_2$ iff they admit the same tangent vector at x .

Remark. If $\sigma : (-\varepsilon, \varepsilon) \rightarrow U$ is a curve at x , $0 < \varepsilon' < \varepsilon$, and $\sigma' = \sigma|_{(-\varepsilon', \varepsilon')} : (-\varepsilon', \varepsilon') \rightarrow U$, then $\sigma \sim \sigma'$.

Definition 6.3. Define the *tangent space* of U at x as $T_x U := \{\sigma : (-\varepsilon, \varepsilon) \rightarrow U \mid \text{smooth curve through } x\} / \sim$.

Remark. We have a natural bijection $\Delta : T_x U \rightarrow \mathbb{R}^n$ taking an equivalence class of curves to the Jacobian at zero. This map is forced to be injective by its definition, and surjectivity is easy to see: take $v \in \mathbb{R}^n$ and $\sigma : (-\varepsilon, \varepsilon) \rightarrow U : t \mapsto x + tv$ (ε is chosen to be small enough so that the image under σ is contained in U). Then $D\sigma|_0 = v$.

Definition 6.4. Let X be an m -dimensional manifold and let $x \in X$. A *curve* in X through x is a smooth function $\sigma : (-\varepsilon, \varepsilon) \rightarrow X$ such that $\sigma(0) = x$.

Let (U, f) be a chart of X at x . If $\varepsilon > 0$ is sufficiently small (we may always assume this), $\sigma(t) \in U$ for all $t \in (-\varepsilon, \varepsilon)$ and we may consider

$$\tilde{\sigma} := f \circ \sigma : (-\varepsilon, \varepsilon) \rightarrow f(U) \subset \mathbb{R}^n.$$

This is a curve in $f(U)$ passing through $f(x)$. Define the relation between two curves $\sigma_i : (-\varepsilon_i, \varepsilon_i) \rightarrow X$ such that $\sigma_i(0) = x$ for $i = 1, 2$ as $\sigma_1 \sim \sigma_2 \iff f \circ \sigma_1 \sim f \circ \sigma_2$.

Lemma 6.5. This definition of equivalence is independent of the chart chosen.

Proof. Let (U, f) and (U', f') be charts on X such that $x \in U \cap U'$ and assume that $f \circ \sigma_1 \sim f \circ \sigma_2$, namely $D(f \circ \sigma_1)|_0 = D(f \circ \sigma_2)|_0$. The transition function $\phi = f' \circ f^{-1} : f(U \cap U') \rightarrow f'(U \cap U')$ is a diffeomorphism and $f' \circ \sigma_i = \phi \circ f \circ \sigma_i$ for $i = 1, 2$. Thus by chain rule we have:

$$D(f' \circ \sigma_1)|_0 = D(\phi \circ f \circ \sigma_1)|_0 = D\phi|_{f(x)} \circ D(f \circ \sigma_1)|_0 = D\phi|_{f(x)} \circ D(f \circ \sigma_2)|_0 = D(\phi \circ f \circ \sigma_2)|_0 = D(f' \circ \sigma_2)|_0$$

and this is just saying $f' \circ \sigma_1 \sim f' \circ \sigma_2$.

Definition 6.6. Let X be a manifold and let $x \in X$. The *tangent space* of X at x is $T_x X = \{\sigma : (-\varepsilon, \varepsilon) \rightarrow X \mid \text{curve through } x\} / \sim$. The elements in $T_x X$ are called *tangent vectors* of X at x .

Proposition 6.7. Let X be an n -dimensional manifold and $x \in X$. Then the tangent space $T_x X$ of X at x is an n -dimensional vector space.

Proof. Recall that $T_x X = \{\sigma : (-\varepsilon, \varepsilon) \rightarrow X \mid \text{curve through } x\} / \sim$. Let (U, f) be a chart of X at x , then we have a bijection

$$T_x X \rightarrow T_{f(x)} f(U) : [\sigma] \mapsto [f \circ \sigma].$$

Since $f(U)$ is an open subset of $\mathbb{R}^n$, there exists a bijection $T_{f(x)} f(U) \rightarrow \mathbb{R}^n$. Thus, we get a bijection $\Delta_f : T_x X \rightarrow \mathbb{R}^n$. This induces the structure of a vector space on $T_x X$:

$$(1) \text{ If } \alpha \in \mathbb{R} \text{ and } v \in T_x X, \text{ then } \alpha \cdot v := \Delta_f^{-1}(\alpha \Delta_f(v))$$

$$(2) \text{ If } v, v' \in T_x X, \text{ then } v + v' := \Delta_f^{-1}(\Delta_f(v) + \Delta_f(v'))$$

Can easily verify the vector space axioms (but this is trivial since all operations inherits from $\mathbb{R}^n$). Therefore, $T_x X$ is a vector space over $\mathbb{R}$ of dimension n .

Remark 6.8. The map $\Delta_f : T_x X \rightarrow \mathbb{R}^n$ depends on the coordinate chart (U, f) . However, the structure of vector space on $T_x X$ does not depend on the choice of the chart: let (U_1, f_1) and (U_2, f_2) be two charts on X with $x \in U_1 \cap U_2$. We showed that for all $\sigma : (-\varepsilon, \varepsilon) \rightarrow X$ with $\sigma(0) = x$, we have $D(f_2 \circ \sigma)|_0 = D(\phi_{21} \circ f_1 \circ \sigma)|_0 = D\phi_{21}|_{f_1(x)} \circ D(f_1 \circ \sigma)|_0$ where ϕ_{21} is the transition function. This is saying:

$$\Delta_{f_2} = D\phi_{21}|_{f_1(x)} \circ \Delta_{f_1}$$

For $J := D\phi_{21}|_{f_1(x)}$ is an invertible linear map. If $\alpha \in \mathbb{R}$ and $v, v' \in T_x X$, then

$$(1) \Delta_{f_2}^{-1}(\alpha \Delta_{f_2}(v)) = (J \circ \Delta_{f_1})^{-1}(\alpha \cdot J \circ \Delta_{f_1}(v)) = \Delta_{f_1}^{-1}(\alpha \Delta_{f_1}(v))$$

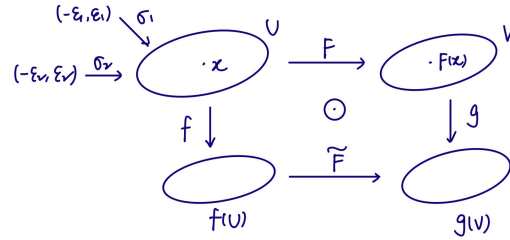
$$(2) \Delta_{f_2}^{-1}(\Delta_{f_2}(v) + \Delta_{f_2}(v')) = (J \circ \Delta_{f_1})^{-1}(J \circ \Delta_{f_1}(v) + J \circ \Delta_{f_1}(v')) = \Delta_{f_1}^{-1}(\Delta_{f_1}(v) + \Delta_{f_1}(v')).$$

Definition 6.9. (Jacobian map of a function between manifolds) Let X and Y be smooth manifolds and let $F : X \rightarrow Y$ be a smooth function. Define the Jacobian map at $x \in X$ as

$$DF|_x : T_x X \rightarrow T_{F(x)} Y : [\sigma] \mapsto [F \circ \sigma].$$

Proposition 6.10. (1) The Jacobian map is well-defined, i.e. for $\sigma_1 \sim \sigma_2$, we have $F \circ \sigma_1 \sim F \circ \sigma_2$.

(2) It's a linear map, which has rank equal to the rank of F at x .



Proof. (1) Let (U, f) be a chart on X with $x \in U$ and (V, g) be a chart on Y with $F(U) \subset V$. By definition of smoothness of F , the map $\tilde{F} = g \circ F \circ f^{-1} : f(U) \rightarrow g(V)$ is smooth. Now let $\sigma_1 : (-\varepsilon_1, \varepsilon_1) \rightarrow X$ and $\sigma_2 : (-\varepsilon_2, \varepsilon_2) \rightarrow X$ be two curves in X through x . We can always assume $\varepsilon_1, \varepsilon_2$ are small enough such that the images of σ_1, σ_2 both lie in U . The condition $\sigma_1 \sim \sigma_2$ tells us that $D(f \circ \sigma_1)|_0 = D(f \circ \sigma_2)|_0$. Now

$$\begin{aligned} D(g \circ F \circ \sigma_1)|_0 &= D(\tilde{F} \circ f \circ \sigma_1)|_0 = D\tilde{F}|_{f(x)} \circ D(f \circ \sigma_1)|_0 = D\tilde{F}|_{f(x)} \circ D(f \circ \sigma_2)|_0 \\ &= D(\tilde{F} \circ f \circ \sigma_2)|_0 = D(g \circ F \circ \sigma_2)|_0. \end{aligned}$$

(2) Firstly notice that for any (equivalence class of) curve σ , $\Delta_g(F \circ \sigma) = D\tilde{F}|_{f(x)} \cdot \Delta_f(\sigma)$

$$DF|_x(\sigma_1 + \sigma_2) = F \circ [\Delta_f^{-1}(\Delta_f(\sigma_1) + \Delta_f(\sigma_2))] = \Delta_g^{-1}(D\tilde{F}|_{f(x)} \cdot (\Delta_f(\sigma_1) + \Delta_f(\sigma_2)))$$

$$= \Delta_g^{-1}(\Delta_g(F \circ \sigma_1) + \Delta_g(F \circ \sigma_2)) = DF|_x(\sigma_1) + DF|_x(\sigma_2)$$

$$DF|_x(a\sigma) = F \circ [\Delta_f^{-1}(a\Delta_f(\sigma))] = \Delta_g^{-1}[D\tilde{F}|_{f(x)} \cdot a\Delta_f(\sigma)] = \Delta_g^{-1}(a\Delta_g(\sigma)) = a \cdot DF|_x(\sigma)$$

So $DF|_x$ is indeed a linear map. Equality in rank can be read off from the following diagram.

$$\begin{array}{ccc}
T_z X & \xrightarrow{\text{Jacobian map } DF|_z} & T_{f(z)} Y \\
\Delta_f \downarrow \wr & \odot & \downarrow \wr \Delta_g \\
\mathbb{R}^n & \xrightarrow{\cdot DF|_{f(z)}} & \mathbb{R}^m
\end{array}$$

Remark 6.11. (Tangent space to submanifolds) Let X be an n -dimensional manifold and let $Z \subset X$ be an m -dimensional submanifold. We showed that $i : Z \hookrightarrow X$ is a smooth immersion, i.e. the function i is of rank m at every $z \in Z$. It follows from 6.10 that $Di|_z$ is a linear map of rank m . Since $T_z Z$ is an m -dimensional vector space, it follows that $Di|_z$ is a linear injection from $T_z Z \rightarrow T_z X$. Consequently, we can write $T_z Z \subset T_z X$ is an m -dimensional subspace of $T_z X$.

Now consider the submanifold Z . By definition we have a chart (U, f) on X with $z \in U$ and $f(U \cap Z) = f(U) \cap A$ for some m -dimensional affine subspace of $\mathbb{R}^n$. Thus, the isomorphism $\Delta_f : T_z X \rightarrow \mathbb{R}^n$ is such that $\Delta_f(T_z Z) = A \subset \mathbb{R}^n$ by looking at the following diagram. (In this procedure we identify the tangent spaces of submanifolds as a subspace of the original tangent space)

$$\begin{array}{ccccc}
& & \Delta_f & & \\
& & \curvearrowright & & \\
T_z Z & \xrightarrow{Di|_z} & T_z X & \xrightarrow{\sim} & T_{f(z)}(f(U)) \xrightarrow{\sim} \mathbb{R}^n \\
& \uparrow & \searrow \text{fun}(Z) = f(U) \cap A & \downarrow \text{UI} & \\
& \text{m-diml} & & & \\
& \text{Subspace} & & & \\
& \sigma & \xrightarrow{\quad} & \sigma & \xrightarrow{\quad} & f_* \sigma
\end{array}$$

Lemma 6.12. Let X be an n -dimensional manifold and let Y be a k -dimensional manifold. Let $F : X \rightarrow Y$ be a smooth function and let $y \in Y$ be a regular value. Denote $Z = F^{-1}(y)$ and $z \in Z$. Then $T_z Z$ is the kernel of the linear map $DF|_z : T_z X \rightarrow T_y Y$.

Proof. By Implicit function theorem, there exists a chart (U, f) of X at z and a chart (V, g) of Y at y such that $F(U) \subset V$ and the function $\tilde{F} = g \circ F \circ f^{-1}$ is the restriction of the standard projection $\pi : \mathbb{R}^n \rightarrow \mathbb{R}^k$. (in the expression of $\tilde{F}$ we omit composition with another diffeomorphism since this diffeomorphism can be absorbed into the chart f)

$$\begin{array}{ccc}
(-\epsilon, \epsilon) \xrightarrow{\sigma} & \begin{array}{c} U \\ \cdot z \end{array} & \xrightarrow{F} \begin{array}{c} V \\ \cdot y \end{array} \\
& \downarrow f & \downarrow g \\
& \begin{array}{c} \text{ } \\ f(U) \subset \mathbb{R}^n \end{array} & \xrightarrow{\tilde{F} = \pi} \begin{array}{c} \text{ } \\ g(V) \subset \mathbb{R}^k \end{array}
\end{array}$$

Now we can identify $DF|_z = \pi : T_z X \rightarrow T_y Y$ due to the following diagram:

$$\begin{array}{ccc}
\mathbb{R}^n & \xrightarrow{\pi} & \mathbb{R}^k \\
\Delta_f \wr \uparrow & & \uparrow \wr \Delta_g \\
T_z X & \xrightarrow{DF|_z} & T_y Y
\end{array}$$

The map from $\mathbb{R}^n \rightarrow \mathbb{R}^k$ is π since the Jacobian of a linear map is itself: $\Delta_g(F \circ \sigma) = D\tilde{F}|_{f(z)} \cdot \Delta_f(\sigma)$. Since $Z = F^{-1}(y)$, $f(Z \cap U) = \tilde{F}^{-1}(g(y)) = f(U) \cap A$ for the affine subspace $A = \pi^{-1}(g(y)) \subset \mathbb{R}^n$.

Therefore, by $\Delta_f(T_z Z) = A$ and $A \cong \ker(\pi)$ by translation,

$$T_z Z \cong A \cong \ker \pi = \ker(T_z X \rightarrow T_y Y)$$

Hence $\ker(T_z X \rightarrow T_y Y) \cong T_z Z$. But we also have $T_z Z \subset \ker(T_z X \rightarrow T_y Y)$, so equality holds.

Example 6.13. (Sphere) Consider $S^n \subset \mathbb{R}^{n+1}$. Let $h : \mathbb{R}^{n+1} \rightarrow \mathbb{R} : (x_0, \dots, x_n) \mapsto \sum x_i^2$. The Jacobian of h at $(x_0, \dots, x_n)$ is $(2x_0, \dots, 2x_n)$, thus any $\alpha \in \mathbb{R} \setminus \{0\}$ is a regular value. In particular, $S^n = h^{-1}(1)$ and if $x = (x_0, \dots, x_n) \in S^n$, then

$$T_x S^n = \ker Dh|_x = \{v \mid x \cdot v = 0\} \subset \mathbb{R}^{n+1}.$$

(Notice that in this case the source and target of h are both Euclidean, consequently the Jacobian map can be identified as the usual Jacobian, by CW3 Q3.)

Example 6.14. (Special Linear Group) Consider an n^2 -dimensional manifold $\mathrm{GL}(n, \mathbb{R})$ and the smooth function $F : \mathrm{GL}(n, \mathbb{R}) \rightarrow \mathbb{R} : A \mapsto \det A$. Clearly $\mathrm{SL}(n, \mathbb{R})$ is a subgroup of $\mathrm{GL}(n, \mathbb{R})$. For $A = (a_{ij})$, $F(A) = \det A = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det S_{ij}$ where S_{ij} is the i, j -minor of A . Now:

$$\frac{\partial F}{\partial a_{ij}}|_A = (-1)^{i+j} \det S_{ij}$$

1 is obviously a regular value since if $\det A = 1$, at least one of the $\det S_{ij}$'s is nonzero, so $\mathrm{SL}(n, \mathbb{R})$ is a submanifold of $\mathrm{GL}(n, \mathbb{R})$ of dimension $n^2 - 1$. The multiplication and inverse functions on $\mathrm{SL}(n, \mathbb{R})$ are still smooth since we can consider the composition of smooth maps

$$\mathrm{SL}(n, \mathbb{R}) \times \mathrm{SL}(n, \mathbb{R}) \xrightarrow{i} \mathrm{GL}(n, \mathbb{R}) \times \mathrm{GL}(n, \mathbb{R}) \xrightarrow[\text{inverse}]{\text{multiplication}} \mathrm{GL}(n, \mathbb{R})$$

Hence $\mathrm{SL}(n, \mathbb{R})$ is itself a Lie Group. We finally compute the tangent space at the point A :

$$T_A \mathrm{SL}(n, \mathbb{R}) = \ker DF|_A = \left\{ x : x \cdot \frac{\partial F}{\partial a_{ij}}|_A = 0 \right\} \subset \mathbb{R}^{n^2}.$$

7 Tangent Bundles, Vector Fields and Flows

Definition 7.1. (special case in $\mathbb{R}^n$) Let $U \subset \mathbb{R}^n$ be an open subset. Define the *tangent bundle* of U as

$$TU := \bigsqcup_{x \in U} T_x U \cong U \times \mathbb{R}^n$$

In particular, $T\mathbb{R}^n = \mathbb{R}^n \times \mathbb{R}^n$. As a product of two n -dimensional manifolds, TU is a manifold of dimension $2n$. It admits a function $\pi : TU \rightarrow U : (x, v) \mapsto x$. A *vector field* on U is a smooth function $\xi : U \rightarrow TU$ such that $\pi \circ \xi = \mathrm{id}_U$. Namely for all $x \in U$, $\xi(x) = (x, v(x))$ where $v(x) \in T_x U$.

Moral. The vector field is just to assign each point in U a vector that is tangent to the manifold in such a way that this is a smooth identification.

Definition 7.2. Let X be a manifold. Define the *tangent bundle* TX of X as the disjoint union of all tangent spaces over X , i.e. $TX := \bigsqcup_{x \in X} T_x X$. There is a projection $\pi : TX \rightarrow X : T_x X \ni v \mapsto x$.

Theorem 7.3. Let X be an n -dimensional manifold. Then the tangent bundle TX is a $2n$ -dimensional manifold and the projection $\pi : TX \rightarrow X$ is smooth.

Proof. Step 1: Define a topology on TX .

Let (U, f) be a chart on X and consider $TU := \pi^{-1}(U) = \bigsqcup_{x \in U} T_x X$. For any $x \in U$, there is a linear isomorphism $\Delta_f : T_x X \rightarrow \mathbb{R}^n$. Thus, we have a bijection

$$F_U : TU = \bigsqcup_{x \in U} T_x X \rightarrow f(U) \times \mathbb{R}^n : T_x X \ni v \mapsto (f(x), \Delta_f(v))$$

A subset $W \subset TU$ is said to be open if $F_U(W)$ is open in $f(U) \times \mathbb{R}^n$ (let F_U induce a topology on TU). We now consider an atlas $\mathcal{A} = \{(U_i, f_i)\}_{i \in I}$ on X and we define the topology on $TU_i = \pi^{-1}(U_i)$ as above for each $i \in I$. A subset $W \subset TX$ is open if $W \cap TU_i$ is open for all $i \in I$. Notice here that $TX = \bigcup_{i \in I} TU_i$, which is not a disjoint union, so there might be overlaps. Also, for any open set $U \subset X$, TU is open in TX since for any $j \in I$, $F_{U_j}(TU \cap TU_j) = F_{U_j}(\bigsqcup_{x \in U \cap U_j} T_x X) = f_j(U \cap U_j) \times \mathbb{R}^n$, which is clearly open.

Step 2: Verify that TX is a topological space which is Hausdorff and second countable. (negligible)

We first check axioms for TU : $\emptyset$ is clearly open in TU , and TU is open in TU since $F_U(TU) = F(U) \times \mathbb{R}^n$. If $W_1, W_2 \subset TU$ are open, then $F_U(W_1) \cap F_U(W_2) = F_U(W_1 \cap W_2)$ is open. If $W_i \subset TU$ is open for any $i \in I$, then $F_U(\bigcup_{i \in I} W_i) = \bigcup_{i \in I} F_U(W_i)$ is open. Then we check axioms for TX : clearly $\emptyset, TX$ are open in TX . If W_1, W_2 open in TX , then $W_1 \cap W_2 \cap TU_i = (W_1 \cap TU_i) \cap (W_2 \cap TU_i)$ open in TU_i for all i . If W_j is open in TX for all $j \in I$, then $(\bigcup_{j \in I} W_j) \cap TU_i = \bigcup_{j \in I} (W_j \cap TU_i)$ open in TU_i for all i .

To show Hausdorffness, take two points $v_1 \neq v_2 \in TX$. If $\pi(v_1) = z_1 \neq \pi(v_2) = z_2$, then since X is Hausdorff, we get open sets U_1, U_2 in X s.t. $U_1 \cap U_2 = \emptyset$, $z_1 \in U_1$, $z_2 \in U_2$. Then TU_1 and TU_2 are two disjoint open sets in TX , they contain v_1 and v_2 respectively. If $\pi(v_1) = \pi(v_2) = z$, we take a chart (U, f) on X with $z \in U$. Since $\mathbb{R}^n$ is Hausdorff, we take two disjoint open sets V_1, V_2 containing $\Delta_f(v_1), \Delta_f(v_2)$ respectively. The subsets $F_U^{-1}(f(U) \times V_1)$ and $F_U^{-1}(f(U) \times V_2)$ of TX are disjoint, and they are open since for any $i \in I$ (I is the index set of the atlas), $F_{U_i}(F_U^{-1}(f(U) \times V_1) \cap TU_i) = f_i(U \cap U_i) \times (\Delta_{f_i} \circ \Delta_f^{-1}(V_1))$ is open.

To show that TX is second-countable, we consider a generic open set $W \subset TX$. By definition $W \cap TU_i$ is an open set in TU_i . Since X is second-countable, countably many TU_i form a cover of TX . But TU_i is also second-countable since it is isomorphic to a subset of $\mathbb{R}^{2n}$, hence TX has a countable basis.

Step 3: Show that TX is a smooth manifold of dimension $2n$.

Since U_i is open in X , TU_i is open. The function

$$F_{U_i} : TU_i = \bigsqcup_{x \in U_i} T_x X \rightarrow f_i(U_i) \times \mathbb{R}^n : T_x X \ni v \mapsto (f_i(x), \Delta_{f_i}(v))$$

is a bijection, and F_{U_i} is in fact a homeomorphism since we defined the topology on TU_i via this map F_{U_i} . At the same time, $f_i(U_i) \times \mathbb{R}^n$ is an open subset of $\mathbb{R}^{2n}$. Hence we conclude that (TU_i, F_{U_i}) is a chart on TX and TX is a topological manifold of dimension $2n$. It remains to show the smoothness of the manifold: for any $i, j \in I$, since X is a manifold, the transition function $\phi_{ij} = f_i \circ f_j^{-1} : f_j(U_i \cap U_j) \rightarrow f_i(U_i \cap U_j)$ is a diffeomorphism. $F_{U_i}(TU_i \cap TU_j) = f_i(U_i \cap U_j) \times \mathbb{R}^n$ and $F_{U_j}(TU_i \cap TU_j) = f_j(U_i \cap U_j) \times \mathbb{R}^n$. Moreover, for any $z \in U_i \cap U_j$, the linear isomorphisms $\Delta_{f_i} : T_z X \rightarrow \mathbb{R}^n$ and $\Delta_{f_j} : T_z X \rightarrow \mathbb{R}^n$ satisfy the relation $\Delta_{f_i} = D\phi_{ij}|_{f_j(z)} \circ \Delta_{f_j}$. Consequently, our transition function for (TU_i, F_{U_i}) and (TU_j, F_{U_j}) is:

$$\Phi_{ij} = F_{U_i} \circ F_{U_j}^{-1} : f_j(U_i \cap U_j) \times \mathbb{R}^n \rightarrow f_i(U_i \cap U_j) \times \mathbb{R}^n : (x, v) \mapsto (\phi_{ij}(x), D\phi_{ij}|_x(v))$$

Φ_{ij} is smooth since the first component is smooth and the second component is linear in v , differentiable w.r.t. x since the derivative of a smooth function is still smooth. By symmetry Φ_{ij} is a diffeomorphism.

Step 4: Show that $\pi : TX \rightarrow X$ is indeed a smooth function (and moreover, a submersion).

Take any $v \in TX$, there exists $i \in I$ such that $v \in TU_i$. Since $F_{U_i} : T_x X \ni v \mapsto (f_i(x), \Delta_{f_i}(v))$, we have

$$\tilde{\pi} = f_i \circ \pi \circ F_{U_i}^{-1} : f_i(U_i) \times \mathbb{R}^n \rightarrow f_i(U_i) : (f_i(x), \Delta_{f_i}(v)) \mapsto f_i(x)$$

This is just the projection onto the first factor so it is clearly smooth. Notice that $D\tilde{\pi}|_{F_{U_i}(v)}$ is an $n \times (2n)$ matrix with the left half an identity, which has rank n . Since choice of v is arbitrary, π is a submersion.

Definition 7.4. Let X be a manifold. A *vector field* on X is a smooth function $\xi : X \rightarrow TX$ such that $\pi \circ \xi = \text{id}_X$. Thus for all $x \in X$, $\xi(x) = (x, v(x))$ where $v(x) \in T_x X$.

Lemma 7.5. Let X be a manifold and let $\xi : X \rightarrow TX$ be a function such that $\pi \circ \xi = \text{id}_X$. Then ξ is a smooth vector field iff for every chart (U, f) on X , $F_U \circ \xi|_U \circ f^{-1} : f(U) \rightarrow f(U) \times \mathbb{R}^n$ is smooth.

Proof. Assume that ξ is smooth and that (U, f) is a chart on X . Since $\pi \circ \xi = \text{id}_X$, it follows that $\xi(U) \subset TU$ and $\xi|_U : U \rightarrow TU$ is a smooth function, which implies that $F_U \circ \xi|_U \circ f^{-1} : f(U) \rightarrow f(U) \times \mathbb{R}^n$ is smooth because this is the composition of 3 smooth functions. Conversely, if for any chart (U, f) on X we have $F_U \circ \xi|_U \circ f^{-1}$ smooth, then for any $x \in X$, taking a chart (U, f) s.t. $x \in U$, clearly $\xi(U) \subset TU$, and TU open in TX , smoothness at x follows from definition, and since x is arbitrary, ξ is smooth.

Proposition 7.6. Let X be a manifold of dimension n and let $Z \subset X$ be a submanifold with inclusion $i : Z \hookrightarrow X$. Then the image of $Di : TZ \rightarrow TX$ is a submanifold of TX .

Proof. Take any $w \in TZ$ and let $z = \pi_Z(w)$. Since Z is a submanifold of X , there is a chart (U, f) on X such that $z \in U$ and $f(U \cap Z) = f(U) \cap A$, where A is an affine subspace of $\mathbb{R}^n$. In particular, for any $z \in Z \cap U$, we have $\Delta_f(T_z Z) = A$. Now consider the chart (TU, F_U) on TX . Clearly $w \in TU$. Recall $F_U : TU \rightarrow f(U) \times \mathbb{R}^n : T_x X \ni v \mapsto (f(x), \Delta_f(v))$, we claim:

$$F_U(TU \cap Di(TZ)) = F_U(TU) \cap (A \times A)$$

This is because for any $(\tilde{z}, \tilde{w}) \in TU \cap Di(TZ)$, the first component of the image is $f(\tilde{z})$, which varies over $f(U) \cap A$; now fix $\tilde{z}$, we have $\tilde{w} \in T_{\tilde{z}}U \cap T_{\tilde{z}}Z$, the second component runs over $\Delta_f(T_{\tilde{z}}U \cap T_{\tilde{z}}Z) = \Delta_f(T_{\tilde{z}}U) \cap A$. Since $A \times A$ is an affine subspace of $\mathbb{R}^{2n}$, we have $Di(TZ)$ a submanifold of TX .

Remark. We simplify the notation and write $TZ \subset TX$ if $Z \subset X$ is a submanifold.

Lemma 7.7. Let X be a manifold and $Z \subset X$ be a submanifold. Let $\xi : X \rightarrow TX$ be a smooth vector field such that $\xi(z) \in T_z Z$ for all $z \in Z$. Then the restriction of ξ to Z is a smooth vector field on Z .

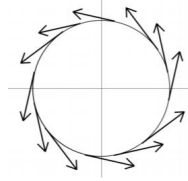
Proof. Since $Z \subset X$ is a submanifold, we have that the inclusion $i : Z \hookrightarrow X$ is a smooth morphism. Hence the restriction $\xi|_Z = \xi \circ i : Z \rightarrow TX$ is smooth. By assumption $\xi|_Z(z) \in TZ$ for all $z \in Z$, thus the induced morphism $\xi|_Z : Z \rightarrow TZ$ is also smooth (we showed before that in the image of a smooth map all lie in a submanifold, then the morphism with codomain restricted to the submanifold is still smooth). Since $\pi \circ \xi = \text{id}_X$, we have $\pi_Z \circ \xi|_Z = \text{id}_Z$ where $\pi_X : TX \rightarrow X$ and $\pi_Z : TZ \rightarrow Z$ are projections. Thus $\xi|_Z$ is a smooth vector field on Z .

Example 7.8. Let $S^1 = \{(x, y) \mid x^2 + y^2 = 1\} \subset \mathbb{R}^2$.

(1) By example 6.13 we know that $T_{(x,y)}S^1 = \{(u, v) \mid ux + vy = 0\} \subset \mathbb{R}^2$.

(2) $\xi : \mathbb{R}^2 \rightarrow T\mathbb{R}^2 : (x, y) \mapsto (-y, x)$ is smooth.

For every $(x, y) \in S^1$, we have $\xi(x, y) \in T_{(x,y)}S^1$. Hence by lemma 7.7, $\xi : S^1 \rightarrow TS^1$ is a vector field on S^1 .



Theorem 7.9. (Hairy Ball) Any vector field on S^2 is equal to zero at some point on S^2 .

Inspiration 7.10. The circle S^1 is a Lie group. We can think of $S^1 \subset \mathbb{R}^2 \cong \mathbb{C}$ and if $z_1, z_2 \in S^1$, their product $z_1 z_2$ and inverses z_1^{-1}, z_2^{-1} are also in S^1 . Multiplication $S^1 \times S^1 \rightarrow S^1$ and inverse $S^1 \rightarrow S^1$ are both smooth since they are smooth functions on $\mathbb{R}^2 \setminus \{0\}$ and S^1 is a submanifold of $\mathbb{R}^2 \setminus \{0\}$.

In polar coordinates, we write $(\cos \theta_1, \sin \theta_1) \cdot (\cos \theta_2, \sin \theta_2) = (\cos(\theta_1 + \theta_2), \sin(\theta_1 + \theta_2))$, and $(\cos \theta, \sin \theta)^{-1} = (\cos(-\theta), \sin(-\theta))$. In particular, for all $s \in \mathbb{R}$, we have a diffeomorphism $F_s : S^1 \rightarrow S^1 : (\cos \theta, \sin \theta) \mapsto (\cos(\theta + 2\pi s), \sin(\theta + 2\pi s))$ such that for all $s, t \in \mathbb{R}$, we have $F_0 = \text{id}_{S^1}$, $F_s \circ F_t = F_{s+t}$. Moreover, F_s defines a smooth function $F : \mathbb{R} \times S^1 \rightarrow S^1 : (s, x) \mapsto F_s(x)$.

Definition 7.11. Let X be a manifold. A *one parameter family of diffeomorphisms* or *flow* on X is a smooth function $F : (-\varepsilon, \varepsilon) \times X \rightarrow X$ for some $\varepsilon > 0$ such that

- (1) for each $s \in (-\varepsilon, \varepsilon)$, the function $F_s : X \rightarrow X : x \mapsto F(s, x)$ is a diffeomorphism;
- (2) $F_0 = \text{id}_X$;
- (3) $F_t \circ F_s = F_{s+t}$ for all $s, t \in (-\varepsilon, \varepsilon)$ such that $s + t \in (-\varepsilon, \varepsilon)$.

Definition 7.12. Let X be a manifold and $F : (-\varepsilon, \varepsilon) \times X \rightarrow X$ be a flow on X . Then for all $x \in X$, we can define $\sigma_x : (-\varepsilon, \varepsilon) \rightarrow X : \sigma_x(t) = F(t, x)$. We have $\sigma_x(0) = F_0(x) = x$, thus σ_x is a curve through x and the equivalence class of σ_x defines a vector $[\sigma_x] \in T_x X$. The function $X \rightarrow TX : x \mapsto [\sigma_x]$ is a vector field on X , called the *infinitesimal version* of the flow F , denoted ξ^F .

Theorem 7.13. Let X be a compact manifold and let ξ be a vector field on X . Then for any $x \in X$, there exists an open subset $U \subset X$ containing x and a flow $F : (-\varepsilon, \varepsilon) \times U \rightarrow X$ such that the restriction $\xi|_U$ is the infinitesimal version of F , i.e. $\xi^F = \xi|_U$.

Remark. Intuition: Starting point in U but might get outwards into X . The proof of this theorem requires results in PDE and the idea is to take chart and then convert this into a problem in $\mathbb{R}^n$.

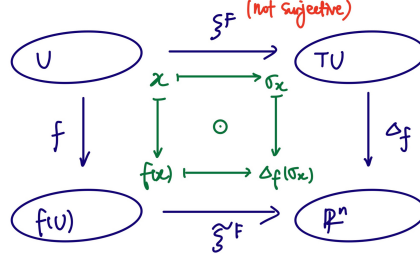
Remark 7.14. Let X be an n -dimensional manifold and let $F : (-\varepsilon, \varepsilon) \times X \rightarrow X$ be a flow on X . Let (U, f) be a chart on X and assume that for all $s \in (-\varepsilon, \varepsilon)$ we have $F_s(U) \subset U$, then we can consider the flow on $f(U)$:

$$\tilde{F} : (-\varepsilon, \varepsilon) \times f(U) \rightarrow f(U) : (t, y) \mapsto f \circ F_t \circ f^{-1}(y)$$

$\tilde{F}$ is smooth since this is the definition of smoothness of F . Since $f(U) \subset \mathbb{R}^n$, we may write $\tilde{F} = (\tilde{F}_1, \dots, \tilde{F}_n)$ where $\tilde{F}_i : (-\varepsilon, \varepsilon) \times f(U) \rightarrow \mathbb{R}$ for $i = 1, \dots, n$. The infinitesimal version of $\tilde{F}$ on $f(U)$ is

$$\tilde{\xi}^F = \frac{\partial \tilde{F}}{\partial s}|_{s=0} = \left(\frac{\partial \tilde{F}_1}{\partial s}|_{s=0}, \dots, \frac{\partial \tilde{F}_n}{\partial s}|_{s=0} \right) : f(U) \rightarrow \mathbb{R}^n$$

Thus $\xi^F = \Delta_f^{-1} \circ \tilde{\xi}^F \circ f : U \rightarrow TU$ is the corresponding vector field on U :



The diagram above commutes since

$$\tilde{\xi}^F(f(x)) = (\partial \tilde{F} / \partial s)|_{s=0, f(x)} = \partial \tilde{F}(s, f(x)) / \partial s|_{s=0} = D(f(F(s, x)))|_{s=0} = D(f(\sigma_x))|_{s=0} = \Delta_f(\sigma_x).$$

Remark. We must require $F_s(U) \subset U$, otherwise this manifold cannot be lifted to Euclidean space. To prove the previous theorem, we need to go backwards: find a chart to convert ξ^F into $\tilde{\xi}^F$ and then solve the PDE.

Example 7.15. (Flow on the circle) We consider the flow given by $F : \mathbb{R} \times S^1 \rightarrow S^1 : (s, x) \mapsto F_s(x)$ where $F_s : S^1 \rightarrow S^1 : (\cos \theta, \sin \theta) \mapsto (\cos(\theta + 2\pi s), \sin(\theta + 2\pi s))$. Take a chart (U, f) on S^1 where $U = \{(x, y) \in S^1 \mid x > 0\}$ and $f : U \rightarrow (-\pi/2, \pi/2) : (\cos \theta, \sin \theta) \mapsto \theta$. Then for $\tilde{F}_s = f \circ F_s \circ f^{-1}$

$$\tilde{F} : (-\varepsilon, \varepsilon) \times (-\pi/2, \pi/2) \rightarrow (-\pi/2, \pi/2) : (s, \theta) \mapsto \theta + 2\pi s$$

Notice here the first component of $\tilde{F}$ cannot be large since otherwise it would leave $f(U) = (-\pi/2, \pi/2)$.

$$\tilde{\xi}^F : (-\pi/2, \pi/2) \rightarrow T(-\pi/2, \pi/2) = (-\pi/2, \pi/2) \times \mathbb{R} : \theta \mapsto (\theta, 2\pi)$$

We have $\xi^F = \Delta_f^{-1} \circ \tilde{\xi}^F \circ f : (\cos \theta, \sin \theta) \mapsto \Delta_f^{-1}(\theta, 2\pi) = (-2\pi \sin \theta, 2\pi \cos \theta) \in T_{(\cos \theta, \sin \theta)} S^1 \subset \mathbb{R}^2$.

Remark. Here Δ_f indeed stands for F_U ; and we can view $(-\sin \theta, \cos \theta)$ be the vector sent to 1 via Δ_f . (this is just a definition or choice: let the map Δ_f take $(-\sin \theta, \cos \theta)$ to 1).

8 Cotangent Spaces

Definition 8.1. Let X be a manifold and let $x \in X$, we denote $R_x(X) = \{h : X \rightarrow \mathbb{R} \mid \text{the rank of } h \text{ at } x \text{ is zero}\} \subset C^\infty(X)$. Define the *cotangent space* of X at x to be the quotient space $T_x^*X = C^\infty(X)/R_x(X)$. Therefore, $T_x^*X = \{[w] \mid w \in C^\infty(X)\}$ where $[w_1] = [w_2]$ if and only if $w_2 - w_1 \in R_x(X)$.

Lemma 8.2. Let $U \subset \mathbb{R}^n$ be an open subset and $x \in U$. Then T_x^*U is a vector space of dimension n .

Proof. Consider the map $\nabla_x : C^\infty(U) \rightarrow \mathbb{R}^n : h \mapsto Dh|_x$. Then ∇_x is linear and $R_x(U) = \ker \nabla_x$ since the rank of a smooth function $U \rightarrow \mathbb{R}$ is zero means that its image under ∇_x is 0. Thus, by first isomorphism theorem $T_x^*U = \text{Im} \nabla_x$. It remains to show that ∇_x is surjective: let $a = (a_1, \dots, a_n) \in \mathbb{R}^n$ and let $h := \sum a_i x_i$, then $\nabla_x(h) = a$, i.e. ∇_x is surjective.

Definition 8.3. Let $\phi : \mathbb{R} \rightarrow \mathbb{R}$ send x to $e^{-1/x}$ for $x > 0$ and 0 for $x \leq 0$. ϕ is a smooth function such that $\phi^{(k)}(0) = 0$ for all $k \geq 0$. Define $\psi(x) = \frac{\phi(x)}{\phi(x) + \phi(1-x)}$, this is a smooth function since $\phi(x) + \phi(1-x) > 0$ for all x . Meanwhile, ψ has the property that $\psi(x) = 0$ for $x \leq 0$ and $\psi(x) = 1$ for $x \geq 1$. Fix a positive integer n and real numbers $0 < r < r'$. Let $B(0, r) = \{x \in \mathbb{R}^n \mid |x| < r\} \subset \mathbb{R}^n$. Define

$$\tilde{\psi} : \mathbb{R}^n \rightarrow \mathbb{R} : x \mapsto \psi \left(\frac{|x| - r'}{r - r'} \right).$$

This map is smooth since it is clearly smooth outside zero but near zero it is constantly equal to 1. Also, $\tilde{\psi}(x) = 0$ if $x \notin B(0, r')$ and $\tilde{\psi}(x) = 1$ if $x \in B(0, r)$. The function $\tilde{\psi}$ is called a *bump function*.

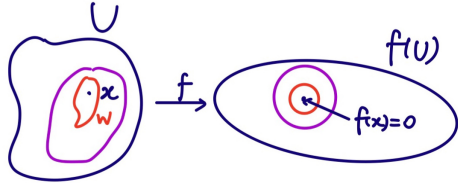
Definition 8.4. Let X be a manifold and let $x \in X$. Let (U, f) be a chart in X s.t. $x \in U$. After translating in $\mathbb{R}^n$, we may assume $f(x) = 0$. Let $r' > 0$ s.t. the closure of $B(0, r')$ is contained in $f(U) \subset \mathbb{R}^n$. Let $0 < r < r'$ and let $\tilde{\psi} : \mathbb{R}^n \rightarrow \mathbb{R}$ be the smooth function we constructed, satisfying $\tilde{\psi}(x) = 0$ if $x \notin B(0, r')$ and $\tilde{\psi}(x) = 1$ for $x \in B(0, r)$. Define

$$\rho : X \rightarrow \mathbb{R} : y \mapsto \begin{cases} (\tilde{\psi} \circ f)(y) & \text{for } y \in U \\ 0 & \text{for } y \notin U \end{cases}$$

Then ρ is a smooth function such that $\rho(y) = 0$ if $y \notin U$ and $\rho = 1$ if $y \in W := f^{-1}(B(0, r))$. We say that ρ is a bump function with respect to the chart (U, f) .

Proposition 8.5. Let X be an n -dimensional manifold and let $x \in X$. Then the cotangent space T_x^*X is a vector space of dimension n .

Proof. Let (U, f) be a chart on X with $x \in U$ and $f(x) = 0$ (this is always plausible by translation). For all $h \in C^\infty(X)$, define $\tilde{h} = h \circ f^{-1} : f(U) \rightarrow \mathbb{R}$. $h \in R_x(X)$ iff the rank of $\tilde{h}$ at $f(x)$ is zero (by definition of rank) iff $D\tilde{h}|_{f(x)} = 0$ iff $h \in \ker \nabla_{f,x}$, where $\nabla_{f,x} : C^\infty(X) \rightarrow \mathbb{R}^n : h \mapsto D\tilde{h}|_{f(x)}$. Therefore, $R_x(X) = \ker \nabla_{f,x}$ and by the first isomorphism theorem we have $T_x^*X = \text{Im } \nabla_{f,x}$. We still need to prove surjectivity. Let $a = (a_1, \dots, a_n) \in \mathbb{R}^n$ and let $\tilde{h} : \mathbb{R}^n \rightarrow \mathbb{R} : x \mapsto \sum a_i x_i$ so that $\nabla_0 \tilde{h} = a$. Let ρ be the bump function w.r.t. the chart (U, f) so that $\rho(y) = 0$ if $y \notin U$ and $\rho = 1$ if $y \in W$, where $W \subset U$ is an open neighbourhood of x . Define $h(y) = \rho \cdot (\tilde{h} \circ f(y))$ if $y \in U$ and $h(y) = 0$ if $y \notin U$. h is a smooth function and inside W we have $\tilde{h} = h \circ f^{-1}$, so $\nabla_{f,x} h = a$. Consequently $T_x^*X \cong \mathbb{R}^n$.



Notation. We will denote $\nabla_{f,x}$ by ∇_f . By abuse of notation, we denote the induced map $\nabla_f : T_x^*X \xrightarrow{\sim} \mathbb{R}^n$. For all $h \in C^\infty(X)$, we denote the equivalence class $dh|_x = [h] \in T_x^*X$.

Remark 8.6. Let $h \in C^\infty(X)$, then $\nabla_f(dh|_x) = D\tilde{h}|_{f(x)}$ for $\tilde{h} = h \circ f^{-1}$; $\nabla_{f'}(dh|_x) = D\tilde{h}'|_{f'(x)}$ where $\tilde{h}' = h \circ f'^{-1}$. Thus $\tilde{h}' = \tilde{h} \circ f \circ f'^{-1} = \tilde{h} \circ \phi^{-1}$. Consequently, we have

$$\nabla_{f'}(dh|_x) = D(\tilde{h} \circ \phi^{-1})|_{f'(x)} = D\tilde{h}|_{f(x)} \circ D\phi^{-1}|_{f'(x)} = \nabla_f(dh|_x) \circ D\phi^{-1}|_{f'(x)}$$

Notice that the input doesn't depend on chart, but the operator depends on chart.

Proposition 8.7. If $U \subset \mathbb{R}^n$ is open and $x \in U$, then $T_x U \cong (T_x^*U)^*$, where $(T_x^*U)^*$ is the dual of T_x^*U .

Proof. Let $v \in T_x U \cong \mathbb{R}^n$. We may write $v = (v_1, \dots, v_n)$. Define the partial derivative at x in the direction v as the linear map

$$\partial_{x,v} : C^\infty(U) \rightarrow \mathbb{R} : h \mapsto Dh|_x(v) = v_1 \frac{\partial h}{\partial x_1} \Big|_x + \dots + v_n \frac{\partial h}{\partial x_n} \Big|_x.$$

Alternatively we consider $v = [\sigma]$ for $\sigma : (-\varepsilon, \varepsilon) \rightarrow U$ being a curve through x , and

$$\partial_{x,v} : C^\infty(U) \rightarrow \mathbb{R} : h \mapsto \frac{d(h \circ \sigma)}{dt} \Big|_0$$

because by chain rule this equals $Dh|_x \cdot D\sigma|_0$. $\partial_{x,v}$ induces a linear map $\partial_{x,v} : T_x^*U \rightarrow \mathbb{R} : dh|_x \mapsto \partial_{x,v}(h)$. This map does not depend on the representative taken in the input since for $h \in R_x(U) = \{h : U \rightarrow \mathbb{R} \mid \text{the rank of } h \text{ is zero}\} \subset C^\infty(U)$, we have $\partial_{x,v}(h) = 0$. Therefore, for any $v \in T_x U$, the map $\partial_{x,v}$ is an element of the dual of T_x^*U (which is defined to be $\text{Hom}(T_x^*U, \mathbb{R})$). We now wish to prove that

$$T_x U \rightarrow (T_x^*U)^* : v \mapsto \partial_{x,v}$$

is a linear isomorphism. (1) This map is clearly linear by construction; (2) This map is injective, since if $v \in T_x U$ is such that $\partial_{x,v} = 0 \in (T_x^*U)^*$, we have $\partial_{x,v}(h) = 0$ for all $dh|_x \in T_x^*U$. If we let $h(y) = v \cdot y$, then

$0 = \partial_{x,v}(h) = v \cdot v$, which implies $v = 0$. Since $\dim T_x U = \dim (T_x^* U)^*$ (for vector space V , $\dim V = \dim V^*$), we conclude that this map is an isomorphism.

Remark. Thus, the tangent space of U at x can be described as the dual of the cotangent space of U at x .

Lemma 8.8. Let V be a vector space of dimension n , then there is a natural isomorphism $\Phi : V \xrightarrow{\sim} (V^*)^*$.

Proof. Let $v \in V$, define $\Phi(v) = \phi_v \in (V^*)^*$ as $\phi_v(\psi) = \psi(v)$ for all $\psi \in V^*$. Let $v \in \ker \Phi$, then $\psi(v) = \Phi(v)(\psi) = 0$ for all $\psi \in V^*$. If $v \neq 0$, by Steinitz Exchange, there exists a basis $\{v_1, \dots, v_{n-1}, v_n = v\}$ of V . Define $\psi : V \rightarrow \mathbb{R}$ by $\psi(\sum a_i v_i) = a_n$ for any $a_1, \dots, a_n \in \mathbb{R}$. Then $\Phi(v)(\psi) = \psi(v) = 1$, thus $v \notin \ker \Phi$, a contradiction. Therefore Φ is injective. Since $\dim V = n = \dim (V^*)^*$, it follows that Φ is an isomorphism.

Corollary. $(T_x U)^* \cong ((T_x^* U)^*)^* \cong T_x^* U$.

Proposition 8.9. Let X be an n -diml manifold and let $x \in X$. Then $T_x X \cong (T_x^* X)^*$ as vector spaces.

Proof. For any $v = [\sigma] \in T_x X$, where $\sigma : (-\varepsilon, \varepsilon) \rightarrow X$ is a curve through x , we define

$$\partial_{x,v} : C^\infty(X) \rightarrow \mathbb{R} : h \mapsto \left. \frac{d(h \circ \sigma)}{dt} \right|_0$$

Let (U, f) be a chart on X with $x \in U$ and consider the curve through $f(x)$, given by

$$\tilde{\sigma} = f \circ \sigma : (-\varepsilon, \varepsilon) \rightarrow f(U) \subset \mathbb{R}^n \quad \tilde{\sigma} = (\tilde{\sigma}_1, \dots, \tilde{\sigma}_n)$$

Then $h \circ \sigma = \tilde{h} \circ \tilde{\sigma}$ where $\tilde{h} = h \circ f^{-1} \in C^\infty(f(U))$. Hence we can write

$$\left. \frac{d(h \circ \sigma)}{dt} \right|_0 = \left. \frac{d(\tilde{h} \circ \tilde{\sigma})}{dt} \right|_0 = D\tilde{h}|_{f(x)} \circ D\tilde{\sigma}|_0 = \sum_{i=1}^n \left. \frac{\partial \tilde{h}}{\partial x_i} \right|_{f(x)} \left. \frac{d\tilde{\sigma}_i}{dt} \right|_0 = \nabla_f (dh|_x)^t \cdot \Delta_f([\sigma])$$

If $h \in R_x(X)$, then $\partial_{x,v}(h) = 0$ since $\nabla_f(dh|_x) = 0$. Hence we get a linear map (abuse of notation):

$$\partial_{x,v} : T_x^* X \rightarrow \mathbb{R} : dh|_x \mapsto \left. \frac{d(h \circ \sigma)}{dt} \right|_0$$

Consequently, $\partial_{x,v} \in (T_x^* X)^*$ and we have a linear map $T_x X \rightarrow (T_x^* X)^* : v \mapsto \partial_{x,v}$.

Recall that $\Delta_f : T_x X \rightarrow \mathbb{R}^n$ and $\nabla_f : T_x^* X \rightarrow \mathbb{R}^n$ are isomorphisms. Consider the chain of $\mathbb{R}$ -linear maps:

$$\begin{array}{ccccccc} \mathbb{R}^n & \xrightarrow{\Delta_f^{-1}} & T_x X & \longrightarrow & (T_x^* X)^* & \xrightarrow{(\nabla_f)^*} & (\mathbb{R}^n)^* \\ \omega & \longmapsto & v = [\sigma] & \longmapsto & \partial_{x,v} & \longmapsto & \tilde{\partial}_{x,v} : \\ & & & & dh|_x \mapsto \nabla_f(dh|_x) \cdot \omega & & \nabla_f(dh|_x) \mapsto \nabla_f(dh|_x) \cdot \omega \end{array}$$

Let $\sigma : (-\varepsilon, \varepsilon) \rightarrow X$ be a curve through x and let $h \in C^\infty(X)$, then if $v = [\sigma]$, we have

$$\partial_{x,v}(dh|_x) = \nabla_f(dh|_x)^t \Delta_f(v)$$

Therefore, the map $\mathbb{R}^n \rightarrow (\mathbb{R}^n)^*$ is given by sending a vector w to the linear map defined by the dot product with w in $\mathbb{R}^n$, this is clearly an isomorphism. As a result, the linear map $T_x X \rightarrow (T_x^* X)^*$ is an isomorphism.

Corollary 8.10. Therefore, the tangent space of X at x is the dual of the cotangent space of X at x , and conversely the cotangent space of X at x is the dual of the tangent space of X at x .

9 Derivations and Vector Bundles

Definition 9.1. Let X be an n -dimensional manifold and let $x \in X$. A *derivation* of X at x is a linear map $\mathfrak{d} : C^\infty(X) \rightarrow \mathbb{R}$ which satisfies the Leibniz rule, i.e. for any $h_1, h_2 \in C^\infty(X)$, we have $\mathfrak{d}(h_1 h_2) = h_1(x) \mathfrak{d}(h_2) + h_2(x) \mathfrak{d}(h_1)$. Denote $\text{Der}_x(X) := \{\text{derivation of } X \text{ at } x\}$.

Lemma 9.2. Let U be an open subset of $\mathbb{R}^n$ containing x and let $\mathfrak{d}$ be a derivation at x . Let $h \in R_x(U)$. Then $\mathfrak{d}(h) = 0$.

Proof. By Leibniz rule, we have $\mathfrak{d}(1) = \mathfrak{d}(1 \cdot 1) = 1 \cdot \mathfrak{d}(1) + 1 \cdot \mathfrak{d}(1) = 2\mathfrak{d}(1)$. Thus $\mathfrak{d}(1) = 0$. Since $\mathfrak{d}$ is a linear function, it follows that $\mathfrak{d}(c) = 0$ for any $c \in \mathbb{R}$. Now let $h \in C^\infty(U)$ be any function. Then Taylor's theorem implies that $h(y) = h(x) + \sum_{i=1}^n (y_i - x_i) \cdot G_i(y)$ for some $G_i \in C^\infty(U)$ (this formulation is usually referred to as the first part of Taylor's thm). Now if $h \in R_x(U)$, then by direct computation we find $G_i(x) = 0$ for any $i = 1, \dots, n$. Therefore:

$$\mathfrak{d}(h) = \sum_{i=1}^n \mathfrak{d}((y_i - x_i) \cdot G_i(y)) = \sum_{i=1}^n ((x_i - x_i) \cdot \mathfrak{d}(G_i(y)) + G_i(x) \cdot \mathfrak{d}(y_i - x_i)) = 0.$$

Lemma 9.3. Let X be a manifold, $x \in X$. Let $\mathfrak{d}$ be a derivation at x . Let U be an open neighbourhood of x in X and let $h \in C^\infty(X)$ be a smooth function such that $h(y) = 0$ for all $y \in U$. Then $\mathfrak{d}(h) = 0$.

Proof. We showed that there exists a bump function $\rho \in C^\infty(X)$ such that there are open sets V, W such that $x \in V \subset W \subset U$ and $\rho|_V = 1$, $\rho|_{X \setminus W} = 0$. Let $\psi = 1 - \rho$, then $\psi \cdot h = h$ since $h = 0$ inside U and $\psi = 1$ outside U . Thus, by Leibniz rule, $\mathfrak{d}(h) = \mathfrak{d}(\psi \cdot h) = h(x)\mathfrak{d}(\psi) + \psi(x)\mathfrak{d}(h) = 0$.

Definition 9.4. Let X be a manifold and let $x \in X$. Consider the set

$$W = \{h \in C^\infty(X) \mid \text{there exists } U \subset X \text{ open s.t. } x \in U \text{ and } h|_U = 0\} \subset C^\infty(X)$$

$W \subset C^\infty(X)$ is a linear subspace and we may consider the quotient $\widehat{C}_x^\infty(X) := C^\infty(X)/W$. The elements of $\widehat{C}_x^\infty(X)$ are called *germs* of smooth functions of X at x . The evaluation map $\widehat{C}_x^\infty(X) \rightarrow \mathbb{R} : [h] \mapsto h(x)$ is well-defined and by lemma 9.3, a derivation $\mathfrak{d} \in \text{Der}_x(X)$ induces a linear map $\widehat{\mathfrak{d}} : \widehat{C}_x^\infty(X) \rightarrow \mathbb{R}$ and it satisfies the Leibniz rule $\widehat{\mathfrak{d}}([h_1] \cdot [h_2]) = h_1(x) \cdot \widehat{\mathfrak{d}}([h_2]) + h_2(x) \cdot \widehat{\mathfrak{d}}([h_1])$.

Lemma 9.5. Let X be a manifold and let $x \in X$. Let $U \subset X$ be an open subset containing x . Then we have an isomorphism $\widehat{C}_x^\infty(X) \xrightarrow{\sim} \widehat{C}_x^\infty(U) : [h] \mapsto [h|_U]$.

Proof. This map is well-defined: for $h \in C^\infty(X)$ a function which is zero in a neighbourhood U_0 of x then its restriction to U is zero in $U \cap U_0$, which is a neighbourhood of x in U . There is a bump function $\rho \in C^\infty(X)$ such that there exist open sets $x \in V \subset W \subset U$ with $\rho|_V = 1$ and $\rho|_{X \setminus W} = 0$. Define the linear map $C^\infty(U) \rightarrow C^\infty(X) : g \mapsto \widehat{g} = \rho \cdot g$ inside U and 0 outside U . If $g \in C^\infty(U)$ is zero in an open neighbourhood U' of U in X then the same holds for $\widehat{g}$. Thus, this map induces a map $\widehat{C}_x^\infty(U) \rightarrow \widehat{C}_x^\infty(X) : [g] \mapsto [\widehat{g}]$. For all $h \in C^\infty(X)$, we have $[\widehat{h|_U}] = h$ on V ; and similarly if $g \in C^\infty(U)$, we have $[\widehat{g}|_U] = g$ on V . Consequently, the induced map $\widehat{C}_x^\infty(U) \rightarrow \widehat{C}_x^\infty(X)$ is the inverse of $\widehat{C}_x^\infty(X) \rightarrow \widehat{C}_x^\infty(U)$, so the claim follows.

Lemma 9.6. Let X be a manifold and let $x \in X$. Let $U \subset X$ be an open subset containing x . Then we have an isomorphism $\text{Der}_x(U) \xrightarrow{\sim} \text{Der}_x(X) : \mathfrak{d}' \mapsto \mathfrak{d}$ where $\mathfrak{d}(h) = \mathfrak{d}'(h|_U)$.

Proof. A derivation $\mathfrak{d}'$ on U induces a map $\widehat{\mathfrak{d}}' : \widehat{C}_x^\infty(U) \rightarrow \mathbb{R}$ which satisfies the Leibniz rule. By lemma 9.5, we have an isomorphism between $\widehat{C}_x^\infty(U)$ and $\widehat{C}_x^\infty(X)$, so we obtain a map $\widehat{\mathfrak{d}} : \widehat{C}_x^\infty(X) \rightarrow \mathbb{R}$. Now, the composition $C^\infty(X) \rightarrow \widehat{C}_x^\infty(X) \rightarrow \mathbb{R}$ also satisfies the Leibniz rule, and this is indeed the map $h \mapsto [h] \mapsto [h|_U] \mapsto \mathfrak{d}'(h|_U) = \mathfrak{d}(h)$. Since this procedure can be reversed, the claim follows.

Theorem 9.7. Let X be a manifold and let $x \in X$. A linear map $\mathfrak{d} : C^\infty(X) \rightarrow \mathbb{R}$ is a derivation if and only if $\mathfrak{d}(h) = 0$ for all $h \in R_x(X)$.

Proof. Assume that $\mathfrak{d} : C^\infty(X) \rightarrow \mathbb{R}$ is a linear map such that $\mathfrak{d}(h) = 0$ for all $h \in R_x(X)$. Let $h_1, h_2 \in C^\infty(X)$ and define $h(y) = h_1(y)h_2(y) - h_1(x)h_2(y) - h_2(x)h_1(y)$. By taking a chart (U, f) with $x \in U$, we can show that $h \circ f^{-1}$ has rank zero, and hence by definition h has rank zero, i.e. $h \in R_x(X)$. Thus, $0 = \mathfrak{d}(h) = \mathfrak{d}(h_1 \cdot h_2) - h_1(x)\mathfrak{d}(h_2) - h_2(x)\mathfrak{d}(h_1)$. It follows that $\mathfrak{d}$ is a derivation.

Conversely, assume that $\mathfrak{d}$ is a derivation and let $h \in R_x(X)$. Let (U, f) be a chart in X with $x \in U$. By lemma 9.6, there is a $\mathfrak{d}' \in \text{Der}_x(U)$ with $\mathfrak{d}(h) = \mathfrak{d}'(h|_U)$. The homeomorphism f induces isomorphisms $C^\infty(U) \rightarrow C^\infty(f(U))$ and $\text{Der}_x(U) \rightarrow \text{Der}_{f(x)}(f(U))$. Let $\tilde{h} := h \circ f^{-1} \in C^\infty(f(U))$. Then $\mathfrak{d}(h) = \mathfrak{d}'(h|_U) = \tilde{\mathfrak{d}}(\tilde{h}) = 0$ due to lemma 9.2 based on the fact that the rank of $\tilde{h}$ at $f(x)$ is 0.

Remark. From this theorem, we can see that a linear map $\mathfrak{d} : C^\infty(X) \rightarrow \mathbb{R}$ is a derivation if and only if it is a function from $C^\infty(X)/R_x(X) \rightarrow \mathbb{R}$, i.e. from $T_x^*(X) \rightarrow \mathbb{R}$. This is saying $\mathfrak{d} \in (T_x^*(X))^* = T_x X$. Thus we get an isomorphism $T_x X \cong \text{Der}_x(X) : v \mapsto \partial_{x,v}$.

Proposition 9.8. Vector fields induce derivations on $\mathbb{R}^n$.

Let $U \subset \mathbb{R}^n$ be an open subset and let $\xi : U \rightarrow TU$ be a vector field. Since $TU = U \times \mathbb{R}^n$, ξ induces a smooth morphism $\tilde{\xi} : U \rightarrow \mathbb{R}^n$. For each $x \in U$, we have a derivative along the direction $\xi(x)$, which is clearly a derivation at x : $\partial_{x, \tilde{\xi}(x)} : C^\infty(U) \rightarrow \mathbb{R}$. For $h \in C^\infty(U)$, we define a map

$$\tilde{\xi}(h) : U \rightarrow \mathbb{R} : x \mapsto \partial_{x, \tilde{\xi}(x)}(h)$$

In coordinates, if we write $\tilde{\xi}(x) = (\tilde{\xi}_1(x), \dots, \tilde{\xi}_n(x))$, then $\tilde{\xi}(h) = \tilde{\xi}_1 \frac{\partial h}{\partial x_1} + \dots + \tilde{\xi}_n \frac{\partial h}{\partial x_n}$. Therefore $\tilde{\xi}(h) \in C^\infty(U)$ and we have a function $\tilde{\xi} : C^\infty(U) \rightarrow C^\infty(U) : h \mapsto \tilde{\xi}(h)$. We may write this as

$$\tilde{\xi} = \sum_{i=1}^n \tilde{\xi}_i \frac{\partial}{\partial x_i}$$

where $\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n}$ are considered as vector fields on U , indeed these are the basis of the set of vector fields on U : for example, $\frac{\partial}{\partial x_1}$ corresponds to $e_1 \in \mathbb{R}^n$.

Now if $h_1, h_2 \in C^\infty(U)$, then the Leibniz rule apply: $\tilde{\xi}(h_1 h_2) = h_1 \tilde{\xi}(h_2) + h_2 \tilde{\xi}(h_1)$. And for this reason, $\tilde{\xi}$ is called a derivation on U . Notice that this is not a derivation at a particular point. Since the Leibniz rule is true for every point of U , it is global!

Definition 9.9. Let X be a manifold. A *derivation* is a linear map $\mathfrak{D} : C^\infty(X) \rightarrow C^\infty(X)$ which satisfies the Leibniz rule $\mathfrak{D}(h_1 h_2) = h_1 \mathfrak{D}(h_2) + h_2 \mathfrak{D}(h_1)$ for all $h_1, h_2 \in C^\infty(X)$. We denote the set of all derivations on X as $\text{Der}(X)$.

Proposition 9.10. Vector fields induce derivations on a manifold.

Let X be an n -dimensional manifold and let $\xi : X \rightarrow TX$ be a vector field. Thus for every $x \in X$, we have a derivation in $\text{Der}_x(X)$, given by the linear map $\partial_{x, \xi(x)} : C^\infty(X) \rightarrow \mathbb{R} : h \mapsto \left. \frac{d(h \circ \sigma)}{dt} \right|_{t=0}$, where $[\sigma]$ is an equivalence class of curves through x , it equals $\xi(x)$. If $h \in C^\infty(X)$, then

$$\xi(h) : X \rightarrow \mathbb{R} : x \mapsto \partial_{x, \xi(x)}(h)$$

We wish to show that $\xi(h)$ is smooth on X . Take $x \in X$ and a chart (U, f) on X and $x \in U$. Let $v = \Delta_f(\xi(x))$ and $\tilde{h} = h \circ f^{-1}$, then we showed in proposition 8.9 that

$$\partial_{x, \xi(x)}(h) = D\tilde{h}|_{f(x)} \circ D\tilde{\sigma}|_0 = \partial_{f(x), v}(\tilde{h}).$$

Let $\tilde{\xi} : f(U) \rightarrow \mathbb{R}^n : y \mapsto \Delta_f(\xi|_{f^{-1}(y)})$ be the induced vector field on $f(U)$. This is smooth by definition of smoothness of ξ . Note that

$$\xi(h) \circ f^{-1} = \tilde{\xi}(\tilde{h})$$

is smooth on $f(U)$ by proposition 9.8 since $f(U)$ is an open subset of $\mathbb{R}^n$. Thus $\xi(h)$ is smooth at x and since this is true for all $x \in X$, $\xi(h) \in C^\infty(X)$. Hence ξ yields a map $C^\infty(X) \rightarrow C^\infty(X) : h \mapsto \xi(h)$ which satisfies the Leibniz rule (since this is just the usual derivative, it automatically satisfies Leibniz). ξ defines a derivation.

Theorem 9.11. Let X be an n -diml manifold. Any derivation $\mathfrak{D} \in \text{Der}(X)$ defines a vector field on X .

Proof. Let $\mathfrak{D} \in \text{Der}(X)$ and let $x \in X$. We have a linear map $\mathfrak{D}|_x : C^\infty(X) \rightarrow \mathbb{R} : h \mapsto (\mathfrak{D}(h))|_x$. The Leibniz rule implies that $\mathfrak{D}|_x \in \text{Der}_x(X)$. But by theorem 9.7, $\text{Der}_x(X) \cong T_x X$. So we define the map

$$\xi : X \rightarrow TX : x \mapsto \mathfrak{D}|_x$$

and we want to show that it is smooth, so that it defines a vector field on X . Let $x \in X$ and take a chart (U, f) on X with $x \in U$, then ξ induces a map $\tilde{\xi} : f(U) \rightarrow f(U) \times \mathbb{R}^n$. We may write $\tilde{\xi}$ as

$$\tilde{\xi} = \sum_{i=1}^n \tilde{\xi}_i \frac{\partial}{\partial x_i} \text{ where } \tilde{\xi}_i : f(U) \rightarrow \mathbb{R} \text{ for } i = 1, \dots, n$$

since $\mathbb{R}^n \cong T_{f(x)} f(U) \cong \text{Der}_{f(x)} f(U)$ where $\frac{\partial}{\partial x_i}$ form a basis of $\text{Der}_{f(x)} f(U)$. Now it suffices to show that $\tilde{\xi}_i$ is smooth at $f(x)$. Let ψ be a bump function such that there exists open sets $f(x) \in W \subset V \subset f(U)$ with $\psi \equiv 1$ on W and $\psi \equiv 0$ outside V . Then we define $\chi_i = (x_i \psi) \circ f$ inside U and 0 outside U . In particular, $\tilde{\chi}_i = \chi_i \circ f^{-1} = x_i$ inside W and by lemma 9.6, we see that for any $y \in f^{-1}(W)$,

$$\mathfrak{D}(\chi_i)|_y = \sum_{j=1}^n \tilde{\xi}_j|_{f(y)} \frac{\partial \tilde{\chi}_i}{\partial x_j} \Big|_{f(y)} = \tilde{\xi}_i|_{f(y)}$$

by identifying $\text{Der}_y(X) \cong \text{Der}_y(U) \cong \text{Der}_{f(y)} f(U)$.

Since $\mathfrak{D}(\chi_i)$ is smooth, $\tilde{\xi}_i$ is also smooth, for $i = 1, \dots, n$ near $f(x)$.

Definition 9.12. Let X be an n -dimensional manifold. A *vector bundle* over X consists of

- (1) a manifold E of dimension $n + r$;
- (2) a smooth surjection $\pi : E \rightarrow X$;
- (3) an atlas $\{(U_i, f_i)\}_{i \in I}$ for X and an atlas $\{(V_i, g_i)\}_{i \in I}$ for E ;

satisfying the following properties:

- (i) $V_i = \pi^{-1}(U_i)$ for all $i \in I$;
- (ii) $g_i(V_i) = f_i(U_i) \times \mathbb{R}^r \subset \mathbb{R}^{n+r}$ for each $i \in I$;
- (iii) $\text{pr}_1 \circ g_i(v) = f_i(\pi(v))$ for all $i \in I$ and $v \in V_i$;
- (iv) $\forall x \in X$, the level set $E_x = \pi^{-1}(x)$ is a vector space of dimension r , such that $\forall i \in I$, the map $g_i|_{E_x} : E_x \rightarrow \mathbb{R}^r$ is an isomorphism.

The number r is called the *rank* of the vector bundle.

Examples 9.13. Let X be an n -dimensional manifold. Then

- (1) The trivial vector bundle $E := X \times \mathbb{R}^r$ is a vector bundle of rank r .
- (2) The tangent bundle is vector bundle of rank n .
- (3) The cotangent bundle $T^*X = \bigsqcup_{x \in X} T_x^*X$ is a vector bundle of rank n .

Proof. The first two are easy. Let's consider (3). Let $T^*X : \bigsqcup_{x \in X} T_x^*X \rightarrow X$ be the cotangent bundle. Select an atlas $\{U_i, f_i\}_{i \in I}$ for X and an atlas $\{(V_i, g_i)\}_{i \in I}$ for T^*X , where $V_i = \bigsqcup_{x \in U_i} T_x^*X$ and $g_i = (f_i, \nabla_{f_i, x})$. We can easily see that this satisfies the definition of a vector bundle.

Definition 9.14. Let X be a manifold and let $\pi : E \rightarrow X$ be a vector bundle. A *section* of E is a smooth morphism $s : X \rightarrow E$ such that $\pi \circ s = \text{id}_X$.

Examples 9.15. Let X be a manifold.

- (1) A section of the tangent bundle TX of X is a vector field on X .
- (2) Let $\pi : E \rightarrow X$ be a vector bundle, then the zero section, defined by mapping every point in X to zero element in E_x : $s : X \rightarrow E : x \mapsto 0 \in E_x$ is a section of E .

Proof. (1) is obvious; for (2), if we consider the corresponding charts (U, f) on X and (V, g) on E with $x \in U$, we may notice that $\tilde{s} : f(U) \rightarrow g(V)$ takes every $f(y) \in f(U)$ to $(f(y), 0)$, and hence s is smooth. $\pi \circ s = \text{id}_X$ is true by definition.

Definition 9.16. Let X be a manifold and let $\pi_1 : E_1 \rightarrow X$ and $\pi_2 : E_2 \rightarrow X$ be two vector bundles over X . An *isomorphism* between E_1 and E_2 is a diffeomorphism $F : E_1 \rightarrow E_2$ such that $\pi_2 \circ F = \pi_1$ and such that the induced function $F_x : (E_1)_x \rightarrow (E_2)_x$ is an isomorphism of vector spaces, for each $x \in X$.

Definition 9.17. Let X be a manifold and let $\pi : E \rightarrow X$ be a vector bundle on X . Then E is *trivial* if it is isomorphic to the trivial vector bundle $X \times \mathbb{R}^r$ for some $r > 0$.

Remark. This is a generalisation of the previous definition and if two vector bundles are isomorphic, their rank would be the same because isomorphism of vector spaces preserve the dimension.

Definition 9.18. Let X be a manifold. A *line bundle* is a vector bundle of rank one.

Lemma 9.19. Let X be a manifold and let $\pi : L \rightarrow X$ be a line bundle on X which admits a section $s : X \rightarrow L$ that is nowhere zero, i.e. $s(x) \neq 0 \in L_x$ for all $x \in X$. Then L is trivial.

Proof. Consider the function $F : X \times \mathbb{R} \rightarrow L : (x, v) \mapsto v \cdot s(x)$. Clearly by definition we have the projection $L \rightarrow X$ composed with F equals the projection $X \times \mathbb{R} \rightarrow L$. Also, for each $x \in X$, since $s(x) \neq 0$, it induces an isomorphism of vector spaces because it sends a vector v to $v \cdot s(x)$ for some fixed $s(x) \neq 0$. We can also see that F is a bijection, so it suffices to show that F and F^{-1} are both smooth.

Since L is a vector bundle, for each $x \in X$ there exists a chart (U, f) on X with $x \in U$ and a chart (V, g) on L such that $V = \pi^{-1}(U)$ and $g(V) = f(U) \times \mathbb{R}$. Then, since s is by definition smooth,

$$\tilde{s} = g \circ s \circ f^{-1} : f(U) \rightarrow f(U) \times \mathbb{R} = g(V)$$

is a smooth function. If $\pi_2 : f(U) \times \mathbb{R} \rightarrow \mathbb{R}$ is the projection into the second factor, we have that

$$\tilde{F} = g \circ F \circ (f^{-1}, \text{id}_{\mathbb{R}}) : f(U) \times \mathbb{R} \rightarrow f(U) \times \mathbb{R} : (y, v) \mapsto (y, v \cdot \pi_2 \circ \tilde{s}(y))$$

This map takes $(y, v) \mapsto (x, v) \mapsto v \cdot s(x) \mapsto v \cdot g(s(x)) = v \cdot \pi_2(\tilde{s}(y))$. Thus, both $\tilde{F}$ and $\tilde{F}^{-1}$ are smooth.

Example 9.20. (Tangent bundle of the circle) Let $S^1 = \{(x, y) \mid x^2 + y^2 = 1\} \subset \mathbb{R}^2$ be the circle. We showed in example 7.8 that S^1 admits a nowhere zero vector field $S^1 \rightarrow TS^1 : (x, y) \mapsto (-y, x)$. Thus the lemma 9.19 says that the tangent bundle of the circle is trivial.

Lemma 9.21. Let X be a manifold and let $E \rightarrow X$ be a trivial vector bundle of rank r . Then E admits a nowhere zero section.

Proof. Let $F : E \rightarrow X \times \mathbb{R}^r$ such that $\pi_2 \circ F = \pi_1$. Consider the function $s : X \rightarrow E : x \mapsto (x, F^{-1}(x, e_1))$. We know that $F \circ s : X \rightarrow X \times \mathbb{R}^r : x \mapsto (x, e_1)$. For any point $x \in X$, take a chart (U, f) with $x \in U$. Then $(f, \text{id}_{\mathbb{R}})$ is a chart on $X \times \mathbb{R}^r$, and $\widetilde{F} \circ s : f(U) \rightarrow f(U) \times \mathbb{R}^r : f(x) \mapsto (f(x), e_1)$, which implies that $F \circ s$ is smooth, and since F is a diffeomorphism, s is smooth. $\pi \circ s = \text{id}_X$ is clear and s is nowhere zero since F^{-1} when fixing x is a linear isomorphism, and $e_1 \neq 0$.

Example 9.22. (Tangent bundle of the sphere) Recall that Hairy Ball Theorem states that if $\xi : S^2 \rightarrow TS^2$ is a vector field, then there exists at least one point x such that $\xi(x) = 0$. It follows that the tangent bundle of the sphere is non-trivial.

10 Cotangent Bundles and One-forms

Definition 10.1. Let X be a manifold. A *one-form* on X is a section of T^*X , i.e. a smooth function $\alpha : X \rightarrow T^*X$ such that $\pi \circ \alpha = \text{id}_X$.

Remark 10.2. (Local description of one-forms) Let X be an n -dimensional manifold and let $x \in X$. Let α be a 1-form on X and let (U, f) be a chart on X such that $x \in U$. Recall that we have the isomorphism $\nabla_f : T_x^*X \xrightarrow{\sim} \mathbb{R}^n$ sending $h \in C^\infty(X)/R_x(X)$ to $D\tilde{h}|_{f(x)}$. Then we get a smooth morphism

$$\tilde{\alpha} : f(U) \rightarrow \mathbb{R}^n : y \mapsto \nabla_f(\alpha|_{f^{-1}(y)})$$

because the following diagram commutes:

$$\begin{array}{ccc} U & \xrightarrow{\alpha} & T_x^*U \\ f \downarrow & & \downarrow \nabla_f \\ f(U) & \xrightarrow{\tilde{\alpha}} & \mathbb{R}^n \end{array}$$

Assume now that (U', f') is also a chart on X with $x \in U'$, let $\phi = f' \circ f^{-1} : f(U \cap U') \rightarrow f'(U \cap U')$ be the transition function. The corresponding one-form on $f'(U')$ is given by $\tilde{\alpha}' : f'(U') \rightarrow \mathbb{R}^n : y \mapsto \nabla_{f'}(\alpha|_{f'^{-1}(y)})$. Recall that for any $h \in C^\infty(X)$, we have the relation

$$\nabla_{f'}(dh|_x) = \nabla_f(dh|_x) \cdot D\phi^{-1}|_{f'(x)}$$

$$\text{and consequently } \tilde{\alpha}'(f'(x)) = \tilde{\alpha}(f(x)) \cdot D\phi^{-1}|_{f'(x)}.$$

Proposition 10.3. (One forms associated to a function on $\mathbb{R}^n$) Let $U \subset \mathbb{R}^n$ be an open subset and let $h \in C^\infty(U)$. Then for all $x \in U$ we have $dh|_x \in T_x^*U = C^\infty(U)/R_x(U)$.

Since we have the isomorphism $T_x^*U \cong \mathbb{R}^n$ (generally, via $\nabla_f(dh|_x) = D\tilde{h}|_{f(x)}$), we may write

$$dh|_x = \left(\frac{\partial h}{\partial x_1} \Big|_x, \dots, \frac{\partial h}{\partial x_n} \Big|_x \right) \in \mathbb{R}^n$$

So we have a smooth one-form on U defined by $dh : U \rightarrow U \times \mathbb{R}^n : x \mapsto (x, dh|_x)$.

In particular, we may consider the coordinate function $x_i : U \rightarrow \mathbb{R}$ and we get the 1-forms dx_i for $i = 1, \dots, n$. Note that dx_i corresponds to the standard vector $(0, \dots, 0, 1, 0, \dots, 0) \in \mathbb{R}^n$, and hence for each $x \in U$, $dx_1(x), \dots, dx_n(x)$ is a basis for T_x^*U . Thus, if α is a one form on U , we may write $\alpha = \alpha_1 dx_1 + \dots + \alpha_n dx_n$ where $\alpha_1, \dots, \alpha_n$ are smooth functions on U (since we are still working in Euclidean space). Notice that we are no longer fixing a point!

For every $h \in C^\infty(U)$ we have

$$dh = \frac{\partial h}{\partial x_1} dx_1 + \dots + \frac{\partial h}{\partial x_n} dx_n$$

This is called an exact one-form.

Proposition 10.4. (One forms associated to a function on a manifold) Let X be a manifold and let $h \in C^\infty(X)$. Then for any $x \in X$, we have $dh : X \rightarrow T^*X : x \mapsto dh|_x$. We want to check that dh is smooth: if (U, f) is a chart on X , we define $\tilde{h} = h \circ f^{-1} \in C^\infty(f(U))$. If (V, g) is the corresponding chart on T^*X , we have that $g \circ dh \circ f^{-1} = d\tilde{h}$, which is a smooth one-form on $f(U)$, thus dh is smooth. (notice here $\tilde{dh}|_{f(x)} = \nabla_f(dh|_x)$ gives the correspondence g).

Example 10.5. Let $X = \mathbb{R}^2$ and let $h \in C^\infty(X)$. Then $dh = \frac{\partial h}{\partial x_1} dx_1 + \frac{\partial h}{\partial x_2} dx_2$. Now consider the 1-form on $\mathbb{R}^2$ defined by $\alpha = \alpha_1 dx_1 + \alpha_2 dx_2$ where α_1, α_2 are smooth functions on $\mathbb{R}^2$. Assume that there exists $h \in C^\infty(X)$ such that $\alpha = dh$, then since dx_1, dx_2 form a basis, $\partial h / \partial x_1 = \alpha_1, \partial h / \partial x_2 = \alpha_2$. Then

$$\frac{\partial \alpha_1}{\partial x_2} = \frac{\partial^2 h}{\partial x_1 \partial x_2} = \frac{\partial \alpha_2}{\partial x_1}$$

Thus, if $\frac{\partial \alpha_1}{\partial x_2} \neq \frac{\partial \alpha_2}{\partial x_1}$, then there does not exist any smooth function h such that $\alpha = dh$. e.g.

- (1) $x_2 dx_1 + x_1 dx_2 = d(x_1 \cdot x_2)$
- (2) $x_2 dx_1 - x_1 dx_2 \neq dh$ for all h .

Example 10.6. Let $X = S^1 = \{(x_1, x_2) \mid x_1^2 + x_2^2 = 1\} \subset \mathbb{R}^2$ and consider $h : S^1 \rightarrow \mathbb{R} : (x_1, x_2) \mapsto x_2^2$. Then dh is a smooth 1-form on X . Let $U = S^1 \cap \{x_1 > 0\}$ and let $f : U \rightarrow \mathbb{R} : (x_1, x_2) \mapsto x_2$. Then (U, f) is a chart on S^1 and we have $\tilde{h} = h \circ f^{-1} : (-1, 1) \rightarrow \mathbb{R} : x_2 \mapsto x_2^2$ and hence $d\tilde{h} = 2x_2 dx_2$.

Now we consider $U' = S^1 \cap \{x_2 > 0\}$ and $f' : U' \rightarrow \mathbb{R} : (x_1, x_2) \mapsto x_1$. Then (U', f') is also a chart on S^1 and we have $\tilde{h}' = h \circ f'^{-1} : (-1, 1) \rightarrow \mathbb{R} : x_1 \mapsto 1 - x_1^2$. Hence $d\tilde{h}' = -2x_1 dx_1$.

Remark 10.7. Recall that if $F : X \rightarrow Y$ is a smooth morphism between manifolds, then there is an induced map $DF|_x : T_x X \rightarrow T_{F(x)} Y$. The cotangent space of X at x (resp. of Y at $F(x)$) is isomorphic to the dual of the tangent space of X at x (resp. of Y at $F(x)$). Recall also that if $L : V \rightarrow W$ is a linear map of vector spaces, then we have an induced map on the dual spaces $L^* : W^* \rightarrow V^* : \psi \mapsto \psi \circ L$. Thus we have

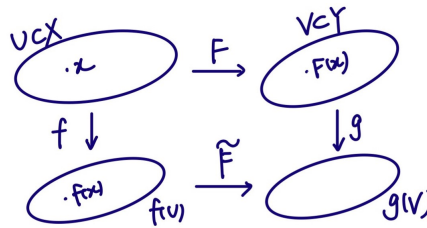
$$DF|_x^* : T_{F(x)}^* Y \rightarrow T_x^* X$$

Let $h \in C^\infty(Y)$, then $h \circ F \in C^\infty(X)$ and we have $DF|_x^*(dh|_{F(x)}) = d(h \circ F)|_x \in T_x^* X$.

This is because $dh|_{F(x)}$ can be thought of as $D(h \circ g^{-1})|_{g(F(x))}$. Then

$$D(h \circ g^{-1})|_{g(F(x))} \circ D\tilde{F}|_{f(x)} = D(h \circ g^{-1} \circ \tilde{F})|_{f(x)} = D(h \circ F \circ f^{-1})|_{f(x)}$$

But the last term corresponds to $d(h \circ F)|_x$.



Definition 10.8. Let X and Y be manifolds and let $F : X \rightarrow Y$ be a smooth morphism. Let α be a one-form on Y , then we have $F^* \alpha : X \rightarrow T^*X : x \mapsto DF|_x^*(\alpha|_{F(x)})$. Once we can show that $F^* \alpha$ is smooth, then it is a one-form on X , called *pull-back* of α .

Remark 10.9. If $U \subset \mathbb{R}^n$ and $V \subset \mathbb{R}^k$ are open subsets, then $F = (F_1, \dots, F_k) : U \rightarrow V$. Let α be a one-form on V , then $F^* \alpha : U \rightarrow U \times \mathbb{R}^n : z \mapsto (DF|_z)^t \cdot \alpha|_{F(z)}$. Thus $F^* \alpha$ is a one-form on U .

Note that this is because $DF|_z^*(\alpha|_{F(z)}) = \alpha|_{F(z)} \circ DF|_z$. So taking transpose on both sides gives the result. (in general, we can say that the map on dual spaces induced by a linear map is multiplying the transpose of the original linear transformation matrix.)

It's easy to see that F^* is linear, i.e. if α_1, α_2 are 1-forms on V and $\lambda_1, \lambda_2 \in \mathbb{R}$, then

$$F^*(\lambda_1 \alpha_1 + \lambda_2 \alpha_2) = \lambda_1 F^* \alpha_1 + \lambda_2 F^* \alpha_2$$

Now let $x_1, \dots, x_n$ be coordinates on U and $y_1, \dots, y_k$ be coordinates on V . There exist smooth functions $\alpha_1, \dots, \alpha_k \in C^\infty(V)$ such that $\alpha = \alpha_1 dy_1 + \dots + \alpha_k dy_k$. Thus, if we consider the expression of $F^*\alpha$, we get

$$F^*\alpha = (\alpha_1 \circ F)(F^*dy_1) + \dots + (\alpha_k \circ F)(F^*dy_k)$$

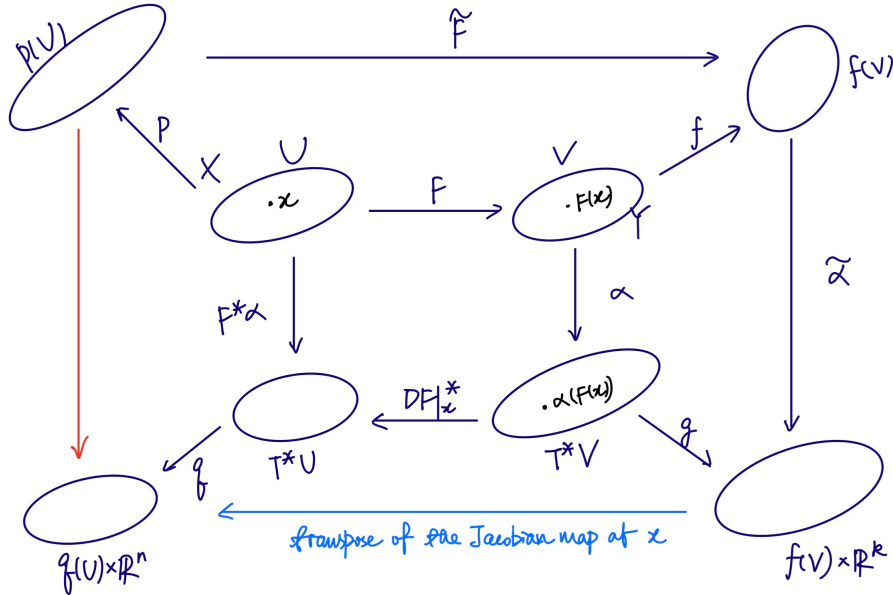
For each $j = 1, \dots, k$, we have

$$(F^*dy_j)|_z = (DF|_z)^t \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} = \left(\frac{\partial F_j}{\partial x_1}, \dots, \frac{\partial F_j}{\partial x_n} \right) \Big|_z$$

Thus we have $F^*dy_j = \frac{\partial F_j}{\partial x_1} dx_1 + \dots + \frac{\partial F_j}{\partial x_n} dx_n$ and hence we expressed $F^*\alpha$ in terms of basis elements.

Proposition 10.10. Let X be an n -dimensional manifold and Y be a k -dimensional manifold. Let $F : X \rightarrow Y$ be a smooth morphism and α be a one-form on Y . Then $F^*\alpha : X \rightarrow T^*X : x \mapsto DF|_x^*(\alpha|_{F(x)})$ is smooth and it defines a 1-form on X . Moreover, if $h \in C^\infty(Y)$, then $F^*dh = d(h \circ F)$.

Proof. Let us consider the following diagram:



Clearly $\tilde{F}$ is smooth since F is smooth, $\tilde{\alpha}$ is smooth since α is a one-form on Y . Since the Jacobian map $DF|_x : T_x U \rightarrow T_{F(x)} V$ can be taken to a linear map $D\tilde{F}|_{p(x)} : T_{p(x)} p(U) \rightarrow T_{\tilde{F}(p(x))} f(V)$, or $\mathbb{R}^n \rightarrow \mathbb{R}^k$, the coefficients in this matrix are smooth functions on $p(U)$. Now since the cotangent spaces are just dual of tangent spaces, the function representing the blue line is essentially left-multiplying by the transpose of the matrix $D\tilde{F}$, all of whose entries are smooth functions on $p(U)$, therefore the red function is smooth, i.e. the pull-back $F^*\alpha$ is smooth. The second claim is already shown in remark 10.7.

11 p -Forms

We first introduce some results in linear algebra.

Definition 11.1. A *multilinear map* on V is a function $\omega : V^p \rightarrow \mathbb{R}$ which is linear on each term, i.e. $\omega(v_1, \dots, v_{k-1}, av_k + a'v'_k, v_{k+1}, \dots, v_p) = a\omega(v_1, \dots, v_k, \dots, v_p) + a'\omega(v_1, \dots, v'_k, \dots, v_p)$ for all $a, a' \in \mathbb{R}$ and any $v_1, \dots, v_p, v'_k \in V$.

Definition 11.2. A multilinear map $\omega : V^p \rightarrow \mathbb{R}$ is called an *alternating p -form* if for all $v_1, \dots, v_p \in V$ and all $\sigma \in S_p$, we have $\omega(v_{\sigma(1)}, \dots, v_{\sigma(p)}) = \epsilon(\sigma)\omega(v_1, \dots, v_p)$. We denote

$$\Lambda^p V^* := \{\omega : V^p \rightarrow \mathbb{R} \mid \omega \text{ is an alternating } p\text{-form}\}$$

Note that $\Lambda^p V^*$ is a vector space, which is called the *p -th exterior power* of V .

Example 11.3. $\Lambda^0 V^* = \mathbb{R}$ (by convention) and $\Lambda^1 V^* = V^* = \text{Hom}(V, \mathbb{R})$.

Definition 11.4. Let $p, q \in \mathbb{Z}_{>0}$. Denote $S_{p,q} := \{\sigma \in S_{p+q} \mid \sigma(1) < \dots < \sigma(p), \sigma(p+1) < \dots < \sigma(p+q)\}$. Given $\omega_1 \in \Lambda^p V^*$ and $\omega_2 \in \Lambda^q V^*$, define the *exterior product* $\omega_1 \wedge \omega_2 \in \Lambda^{p+q} V^*$ by, for each $v_1, \dots, v_{p+q} \in V$,

$$\omega_1 \wedge \omega_2(v_1, \dots, v_{p+q}) := \sum_{\sigma \in S_{p,q}} \epsilon(\sigma) \omega_1(v_{\sigma(1)}, \dots, v_{\sigma(p)}) \omega_2(v_{\sigma(p+1)}, \dots, v_{\sigma(p+q)})$$

Proposition 11.5. Assume that $\omega_1, \omega_2 \in \Lambda^1 V^*$, then $\omega_1 \wedge \omega_2(v, w) = \omega_1(v)\omega_2(w) - \omega_1(w)\omega_2(v)$. In general, if $\omega_1, \dots, \omega_p \in \Lambda^1 V^*$ and $v_1, \dots, v_p \in V$, then

$$\omega_1 \wedge \dots \wedge \omega_p(v_1, \dots, v_p) = \det \begin{bmatrix} \omega_1(v_1) & \omega_1(v_2) & \dots & \omega_1(v_p) \\ \omega_2(v_1) & \omega_2(v_2) & \dots & \omega_2(v_p) \\ \vdots & \vdots & \dots & \vdots \\ \omega_p(v_1) & \omega_p(v_2) & \dots & \omega_p(v_p) \end{bmatrix}$$

Proposition 11.6. Let $\omega_i \in \Lambda^{p_i} V^*$ for $i = 1, 2, 3$. Then the following properties hold:

- (1) Associativity. $\omega_1 \wedge (\omega_2 \wedge \omega_3) = (\omega_1 \wedge \omega_2) \wedge \omega_3$.
- (2) Distributivity. If $p_2 = p_3$, then $\omega_1 \wedge (\omega_2 + \omega_3) = \omega_1 \wedge \omega_2 + \omega_1 \wedge \omega_3$.
- (3) Super-commutativity. $\omega_1 \wedge \omega_2 = (-1)^{p_1 p_2} \omega_2 \wedge \omega_1$.

Definition 11.7. Let $\Phi : V \rightarrow W$ be a linear morphism between vector spaces. Let $\omega \in \Lambda^p W^*$, then the *pull-back* $\Phi^* \omega \in \Lambda^p V^*$ of ω is defined by, for all $v_1, \dots, v_p \in V$,

$$(\Phi^* \omega)(v_1, \dots, v_p) = \omega(\Phi v_1, \dots, \Phi v_p)$$

Notice here $(\Phi^* \omega)(v_{\sigma(1)}, \dots, v_{\sigma(p)}) = \omega(\Phi(v_{\sigma(1)}), \dots, \Phi(v_{\sigma(p)})) = \epsilon(\sigma) \omega(\Phi(v_1), \dots, \Phi(v_p)) = \epsilon(\sigma) (\Phi^* \omega)(v_1, \dots, v_p)$. So our definition is valid since this is an alternating p -form.

Proposition 11.8. Let $\Phi : V \rightarrow W$ and $\Psi : W \rightarrow Z$ be linear maps between vector spaces. Then:

- (1) the pull-back $\Phi^* : \Lambda^p W^* \rightarrow \Lambda^p V^* : \omega \mapsto \Phi^* \omega$ is linear and it preserves the exterior product, i.e. $\Phi^*(\omega_1 \wedge \omega_2) = \Phi^* \omega_1 \wedge \Phi^* \omega_2$.
- (2) $(\Psi \circ \Phi)^* \omega = \Phi^* \Psi^* \omega$
- (3) Assume $V = W$ and $p = \dim V$, then $\Phi^* \omega = \det(\Phi) \omega$.

Proof. (1) $\Phi^*(\omega_1 \wedge \omega_2)(v_1, \dots, v_p) = (\omega_1 \wedge \omega_2)(\Phi v_1, \dots, \Phi v_p) = \sum_{\sigma \in S_{p_1, p_2}} \epsilon(\sigma) \omega_1(\Phi v_{\sigma(1)}, \dots, \Phi v_{\sigma(p_1)}) \omega_2(\Phi v_{\sigma(p_1+1)}, \dots, \Phi v_{\sigma(p)})$
 $= \sum_{\sigma \in S_{p_1, p_2}} \epsilon(\sigma) \Phi^* \omega_1(v_{\sigma(1)}, \dots, v_{\sigma(p_1)}) \Phi^* \omega_2(v_{\sigma(p_1+1)}, \dots, v_{\sigma(p)}) = \Phi^* \omega_1 \wedge \Phi^* \omega_2(v_1, \dots, v_p)$. (2) is a trivial result from definition. (3) is tedious to write down, but this can be easily seen to be the case by writing Φ as a matrix w.r.t. a basis of V and writing each v_i to be a linear combination of basis vectors.

Proposition 11.9. Let V be an n -dimensional vector space.

- (1) If $p > n$ then $\Lambda^p V^* = 0$.
- (2) Let $v_1, \dots, v_n$ be a basis. Let $v_1^*, \dots, v_n^*$ be the dual basis of V^* , i.e.

$$v_i^*(v_j) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$$

Then for all $0 < p \leq n$, the set $v_{i_1}^* \wedge \dots \wedge v_{i_p}^*$ for $i_1 < i_2 < \dots < i_p$ is a basis of $\Lambda^p V^*$. In particular,

$$\dim \Lambda^p V^* = \binom{n}{p}$$

Proof. (1) is easy to see: we know that if $p > n$, then there must be a linear relation between the entry vectors. If two entry vectors are the same, then when applied by ω , the result is zero. So we can always add such terms to construct the zero on a certain coordinate. Hence the output under ω must be zero, i.e. $\Lambda^p V^* = 0$. (2) A generic element ω is a function sending $(u_1, \dots, u_p)$ to some real number for any $u_i \in V$. We may wish to write each u_i in terms of the basis $v_1, \dots, v_n$ and then apply the multilinear property. ω is uniquely determined by how it operates on $(v_{i_1}, \dots, v_{i_p})$ so it is a sum of elements of the form $v_{i_1}^* \wedge \dots \wedge v_{i_p}^*$ for $i_1 < \dots < i_p$. So alternating p -forms in this form form a basis for $\Lambda^p V^*$.

Definition 11.10. Let X be an n -dimensional manifold and let $x \in X$, denote $\Lambda^p T_x^* X := \Lambda^p (T_x X)^*$. Note

that when $p = 1$, $\Lambda^1 T_x^* X = T_x^* X = T_x^* X$. Define $\Lambda^p T^* X := \bigsqcup_{x \in X} \Lambda^p T_x^* X \xrightarrow{\pi} X$. This is called the p -th exterior bundle of X and it is a vector bundle of rank $\mathbf{C}_n^p$. The sections of $\Lambda^p T^* X$ are called *differential p -forms* on X . For ω a differential p -form on X , for any $x \in X$, $\omega(x)$ is an alternating p -form on $T_x X$.

Definition 11.11. Denote $\Omega^p(X) := \{\omega = \text{differential } p\text{-form on } X\}$ and $\Omega^\bullet X := \bigoplus_{p=0}^n \Omega^p(X)$. Notice that here $\Omega^0(X) \cong \{f : X \rightarrow \mathbb{R} \text{ is a smooth morphism}\} = C^\infty(X)$; and if X has dimension n , then $\Omega^p(X) = 0$ for all $p \geq n + 1$.

Definition 11.12. Given $\omega_1 \in \Omega^p(X)$ and $\omega_2 \in \Omega^q(X)$, define $\omega_1 \wedge \omega_2(x) = \omega_1(x) \wedge \omega_2(x) \in \Lambda^{p+q} T_x^* X$. Note that associativity, distributivity and super-commutativity hold and when we vary $x \in X$, $\omega_1 \wedge \omega_2$ is a smooth section of the $(p + q)$ -th exterior bundle, so we write $\omega_1 \wedge \omega_2 \in \Omega^{p+q}(X)$.

Definition 11.13. Let X and Y be manifolds and $F : X \rightarrow Y$ be a smooth morphism. Then for all $x \in X$, we have that the Jacobian of F at x is a linear map $DF|_x : T_x X \rightarrow T_{F(x)} Y$. Define $F^* : \Omega^p(Y) \rightarrow \Omega^p(X)$ by: for any $\omega \in \Omega^p(Y)$, $x \in X$ and $v_1, \dots, v_p \in T_x X$,

$$F^*\omega(x)(v_1, \dots, v_p) = \omega(F(x))(DF|_x v_1, \dots, DF|_x v_p)$$

The differential p -form $F^*\omega : X \rightarrow \Lambda^p T^* X$ is called the *pull-back* of ω .

Proposition 11.14. Let $F : X \rightarrow Y$ be a smooth function between manifolds.

- (1) If $\omega_1 \in \Omega^p(Y)$ and $\omega_2 \in \Omega^q(Y)$, then $F^*(\omega_1 \wedge \omega_2) = F^*(\omega_1) \wedge F^*(\omega_2)$;
- (2) If $G : Y \rightarrow Z$ is a smooth function between manifolds and $\omega \in \Omega^p(Z)$, we have $(G \circ F)^*\omega = F^*G^*\omega$.

Proof. For (1) and (2), when evaluating at any point, this is clear by proposition 11.8 (for the second statement, the linear map behind is just the Jacobian map).

Remark 11.15. (Local description of p -forms) Let $U \subset \mathbb{R}^n$ be an open subset and $x \in U$. Then if we consider the identification of $T_x X \cong \text{Der}_x X$, we have $\left\{\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n}\right\}$ is a basis of $T_x U$ and we get a dual basis $dx_i \in T_x^* U$ by defining $dx_i(\partial/\partial x_j) = \delta_{i,j}$. Then $\{dx_1, \dots, dx_n\}$ is a basis of $T_x^* U$. A basis of $\Lambda^p T_x^* U$ is then given by $\{dx_{i_1} \wedge \dots \wedge dx_{i_p} \mid 1 \leq i_1 < \dots < i_p \leq n\}$. A differential p -form ω on U is locally given by

$$\omega = \sum_{|l|=p} h_l dx_{i_1} \wedge \dots \wedge dx_{i_p} : U \rightarrow \Lambda^p T^* U$$

where $h_l : U \rightarrow \mathbb{R}$ is a C^∞ -function for all $l = (i_1, \dots, i_p)$. Recall that for any $i, j \in \{1, \dots, n\}$, $dx_i \wedge dx_j = -dx_j \wedge dx_i$ and $dx_i \wedge dx_i = 0$, so we just need the strictly increasing indices $i_1 < \dots < i_p$.

Example 11.16. Let $F : U \rightarrow V$ be a smooth function between subsets of $\mathbb{R}^n$. Let ω be an n -form on V . Then locally we may write $\omega = h dy_1 \wedge \dots \wedge dy_n$ where h is a smooth function. Then by proposition 11.8, $F^*\omega(x)(v_1, \dots, v_n) = \omega(F(x))(DF|_x v_1, \dots, DF|_x v_n) = h(F(x)) \cdot (dy_1 \wedge \dots \wedge dy_n|_{F(x)})(DF|_x v_1, \dots, DF|_x v_n) = h(F(x)) \cdot (\det DF|_x) \cdot (dx_1 \wedge \dots \wedge dx_n)(v_1, \dots, v_n)$ (by using a similar idea as in 11.8 (3)). Hence

$$F^*\omega(x) = h \circ F(x) \cdot \det DF|_x \cdot (dx_1 \wedge \dots \wedge dx_n)$$

A very important remark. Notice that v_i 's have coordinates in x_i 's, $DF|_x v_i$'s have coordinates in y_i 's.

Definition 11.17. Let $U \subset \mathbb{R}^n$ be an open subset. Given a smooth function $h \in C^\infty(U)$, the differential of h is locally defined by $dh = \sum_{i=1}^n \frac{\partial h}{\partial x_i} dx_i$, where dh, dx_i are one-forms on U . We may see this in 2 ways:

- (1) $dh(\xi) = \xi(h)$ sending x to $\partial_{x, \xi(x)}(h)$ for any vector field $\xi : U \rightarrow TU$.
- (2) $dh = h^* dx$, where $h^* dx(y)(v) = dx|_{h(y)}(Dh|_y \cdot v) = Dh|_y \cdot v = dh(y)(v)$ for $dh \in \Omega^1(U)$, $dx \in \Omega^1(\mathbb{R})$.

Generally, we know that $\omega \in \Omega^p(U)$ is locally given by $\omega = \sum_{|l|=p} h_l dx_{i_1} \wedge \dots \wedge dx_{i_p}$. Define the *differential* of ω to be $d\omega = \sum_{|l|=p} dh_l \wedge dx_{i_1} \wedge \dots \wedge dx_{i_p}$. Here $d\omega \in \Omega^{p+1}(U)$ is a $(p + 1)$ -form, and the induced map $d : \Omega^p(U) \rightarrow \Omega^{p+1}(U)$ is called the *de Rham differential* on U .

Remarks 11.18. Let X be an n -dimensional manifold and $\pi : E \rightarrow X$ be a vector bundle of rank r . Then there exists an atlas $\{(U_i, f_i)\}_{i \in I}$ for X and an atlas $\{(V_i, g_i)\}_{i \in I}$ for E such that $V_i = \pi^{-1}(U_i)$ and $g_i(V_i) = f_i(U_i) \times \mathbb{R}^r$ for any $i \in I$. If $s : X \rightarrow E$ is a section, then $\pi \circ s = \text{id}_X$ and we may define $\tilde{s}_i := g_i \circ s \circ f_i^{-1} : f_i(U_i) \rightarrow f_i(U_i) \times \mathbb{R}^r$ for any $i \in I$. For any $i, j \in I$ such that $U_i \cap U_j \neq \emptyset$, we have the transition functions $\phi_{ij} : f_j(U_i \cap U_j) \rightarrow f_i(U_i \cap U_j)$ and $\Phi_{ij} : f_j(U_i \cap U_j) \times \mathbb{R}^r \rightarrow f_i(U_i \cap U_j) \times \mathbb{R}^r$. The functions $\tilde{s}_i := g_i \circ s \circ f_i^{-1} : f_i(U_i) \rightarrow f_i(U_i) \times \mathbb{R}^r$ for any $i \in I$ are s.t. the following diagram is commutative:

$$\begin{array}{ccc}
f_j(U_i \cap U_j) & \xrightarrow{\tilde{s}_j|_{f_j(U_i \cap U_j)}} & f_j(U_i \cap U_j) \times \mathbb{R}^r \\
\downarrow \phi_{ij} & & \downarrow \Phi_{ij} \\
f_i(U_i \cap U_j) & \xrightarrow{\tilde{s}_i|_{f_i(U_i \cap U_j)}} & f_i(U_i \cap U_j) \times \mathbb{R}^r
\end{array}$$

i.e. we have $\Phi_{ij} \circ \tilde{s}_j|_{f_j(U_i \cap U_j)} = \tilde{s}_i|_{f_i(U_i \cap U_j)} \circ \phi_{ij}$. We also have the opposite direction: if $\tilde{s}_i : f_i(U_i) \rightarrow f_i(U_i) \times \mathbb{R}$ for any $i \in I$ are functions s.t. $\Phi_{ij} \circ \tilde{s}_j|_{f_j(U_i \cap U_j)} = \tilde{s}_i|_{f_i(U_i \cap U_j)} \circ \phi_{ij}$, then there exists a section $s : X \rightarrow E$ s.t. $\tilde{s} = g_i \circ s \circ f_i^{-1}$ for any $i \in I$.

Proposition 11.19. Let X be an n -dimensional manifold and let $\omega \in \Omega^p(X)$ be a p -form. Let $\{(U_i, f_i)\}_{i \in I}$ be an atlas on X . Then for each $i \in I$, we can consider $\omega_i = (f_i^{-1})^* \omega \in \Omega^p(f_i(U_i))$. Let $d\omega_i \in \Omega^{p+1}(f_i(U_i))$ be its de Rham differential. Then the $d\omega_i$ defines a $(p+1)$ -form $d\omega \in \Omega^{p+1}(X)$. This form $d\omega$ is called the de Rham differential of ω .

Proof. It suffices to show that $d\omega_i = \phi_{ji}^* d\omega_j$. But we have $\omega_i = \phi_{ji}^* \omega_j$. So result follows from 11.20 (3).

Proposition 11.20. For any open subsets $U \subset \mathbb{R}^n$, $V \subset \mathbb{R}^k$,

- (1) (Leibniz rule) If $\omega_1 \in \Omega^p(U)$ and $\omega_2 \in \Omega^q(U)$, then $d(\omega_1 \wedge \omega_2) = d\omega_1 \wedge \omega_2 + (-1)^p \omega_1 \wedge d\omega_2$.
- (2) For all $p \geq 0$, the map $d \circ d : \Omega^p(U) \rightarrow \Omega^{p+2}(U)$ is the zero map, i.e. $d \circ d = 0$.
- (3) If $F : U \rightarrow V$ is a smooth function between manifolds, then for any $\omega \in \Omega^p(V)$, $F^* d\omega = d(F^* \omega)$.

Proof. (1) Assume that $\omega_1 = h_1 dx_{i_1} \wedge \cdots \wedge dx_{i_p}$ and $\omega_2 = h_2 dx_{j_1} \wedge \cdots \wedge dx_{j_q}$ for $h_1, h_2 \in C^\infty(U)$. Then

$$\begin{aligned}
d(\omega_1 \wedge \omega_2) &= d(h_1 h_2 dx_{i_1} \wedge \cdots \wedge dx_{i_p} \wedge dx_{j_1} \wedge \cdots \wedge dx_{j_q}) = (h_2 dh_1 + h_1 dh_2) \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_p} \wedge dx_{j_1} \wedge \cdots \wedge dx_{j_q} \\
&= (dh_1 \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_p}) \wedge (h_2 dx_{j_1} \wedge \cdots \wedge dx_{j_q}) + (-1)^p (h_1 dx_{i_1} \wedge \cdots \wedge dx_{i_p}) \wedge (dh_2 \wedge dx_{j_1} \wedge \cdots \wedge dx_{j_q}) \\
&= d\omega_1 \wedge \omega_2 + (-1)^p \omega_1 \wedge d\omega_2.
\end{aligned}$$

And the general Leibniz rule follows from linearity.

(2) Assume that $\omega = h dx_{i_1} \wedge \cdots \wedge dx_{i_p}$ for $h \in C^\infty(U)$. Then $d\omega = \sum_{j=1}^n (\frac{\partial h}{\partial x_j} dx_j) \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_p}$

$$\implies d(d\omega) = \sum_{m=1}^n \sum_{j=1}^n \frac{\partial^2 h}{\partial x_m \partial x_j} dx_m \wedge dx_j \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_p}.$$

Since $h \in C^\infty(U)$, we have $\frac{\partial^2 h}{\partial x_m \partial x_j} dx_m \wedge dx_j = -\frac{\partial^2 h}{\partial x_j \partial x_m} dx_j \wedge dx_m$, and hence $d(d\omega) = 0$.

By linearity, it follows that $d \circ d = 0$.

(3) Write $\omega = h dy_{i_1} \wedge \cdots \wedge dy_{i_p}$ for $h \in C^\infty(V)$. By proposition 11.14, if $\omega_1 \in \Omega^p(V)$ and $\omega_2 \in \Omega^q(V)$, then $F^*(\omega_1 \wedge \omega_2) = F^*(\omega_1) \wedge F^*(\omega_2)$. Hence $F^*(\omega) = h \circ F \cdot (F^* dy_{i_1}) \wedge \cdots \wedge (F^* dy_{i_p})$. Also, we know from proposition 10.10 that for $g \in C^\infty(V)$, then $F^* dg = d(g \circ F)$. Thus, $d(F^* dy_{i_j}) = d(g \circ F) = 0$. Thus

$$\begin{aligned}
d(F^* \omega) &= d(h \circ F) \wedge (F^* dy_{i_1}) \wedge \cdots \wedge (F^* dy_{i_p}) \text{ by Leibniz rule and } d(F^* dy_{i_j}) = 0 \\
&= F^* dh \wedge (F^* dy_{i_1}) \wedge \cdots \wedge (F^* dy_{i_p}) = F^*(dh \wedge dy_{i_1} \wedge \cdots \wedge dy_{i_p}) \\
&= F^* d\omega
\end{aligned}$$

And the general result follows from linearity.

Remark 11.21. The proposition 11.20 can be generalised for arbitrary manifolds: for manifolds X and Y ,

- (1) (Leibniz rule) If $\omega_1 \in \Omega^p(X)$ and $\omega_2 \in \Omega^q(X)$, then $d(\omega_1 \wedge \omega_2) = d\omega_1 \wedge \omega_2 + (-1)^p \omega_1 \wedge d\omega_2$.
- (2) For all $p \geq 0$, the map $d \circ d : \Omega^p(X) \rightarrow \Omega^{p+2}(X)$ is the zero map, i.e. $d \circ d = 0$.
- (3) If $F : X \rightarrow Y$ is a smooth function between manifolds, then for any $\omega \in \Omega^p(Y)$, $F^* d\omega = d(F^* \omega)$.

This is because all the properties in 11.20 are local so we can just show these for an open neighbourhood of each $x \in X$, so STP the result for every open set in an open cover.

Definition 11.22. Let $\omega \in \Omega^p(X)$ be a differential p -form, then

- (1) ω is said to be closed if $d\omega = 0$;
- (2) ω is said to be exact if there exists $\omega' \in \Omega^{p-1}(X)$ such that $d\omega' = \omega$.

Remark 11.21 implies that any exact p -form is closed but the opposite is not necessarily true.

12 Integration

Recall that if $D \subset \mathbb{R}^n$ is a compact subset with border of measure zero, $h : \mathbb{R}^n \rightarrow \mathbb{R}$ is a smooth function and $\phi : D \rightarrow \phi(D) \subset \mathbb{R}^n$ is a diffeomorphism, then $\int_{\phi(D)} h dy_1 \wedge \cdots \wedge dy_n = \int_D h \circ \phi \cdot |\det d\phi| dx_1 \wedge \cdots \wedge dx_n$. Thus if $\omega \in \Omega^n(D)$ is the n -form given by $\omega = h dx_1 \wedge \cdots \wedge dx_n$, we may define

$$\int_D \omega := \int_D h dx_1 \wedge \cdots \wedge dx_n$$

Definition 12.1. Let $U \subset \mathbb{R}^n$ be an open subset, the *support* of a differential p -form $\omega \in \Omega^p(U)$ is $\text{Supp}(\omega) := \{x \in U \mid \omega(x) \neq 0\}$. If $D = \text{Supp}(\omega)$ is compact and $p = n$, then $\int_U \omega := \int_D \omega$.

Proposition 12.2. If $\phi : V \rightarrow U$ is a diffeomorphism s.t. $\det(D\phi)_x > 0$ for all $x \in V$, then $\int_U \omega = \int_V \phi^* \omega$.

Proof. LHS = $\int_U h dy_1 \wedge \cdots \wedge dy_n$, RHS = $\int_V h \circ \phi(x) \cdot \det(D\phi|_x) dx_1 \wedge \cdots \wedge dx_n$.

Definition 12.3. Let V be a vector space over $\mathbb{R}$ of dimension n and let $B_1 = (v_1, \dots, v_n)$ and $B_2 = (w_1, \dots, w_n)$ be ordered bases of V . Then B_1 and B_2 have the same *orientation* if $\det T > 0$, where $T : V \rightarrow V$ is the linear map defined by $T(v_i) = w_i$ for all $i = 1, \dots, n$.

If we fix $\omega \in \Lambda^n V^*$, then an *orientation* Λ of V is the set of all ordered basis $(v_1, \dots, v_n)$ s.t. $\omega(v_1, \dots, v_n) > 0$.

Remark. If $B_1 = (v_1, \dots, v_n)$ and $B_2 = (w_1, \dots, w_n)$ are ordered bases of V with the same orientation, then $\omega(v_1, \dots, v_n) > 0$ if and only if $\omega(w_1, \dots, w_n) > 0$. This is because if L is the linear transformation taking $v_i \mapsto w_i$, then $L^* \omega(v_1, \dots, v_n) = \det(L) \cdot \omega(v_1, \dots, v_n) = \omega(w_1, \dots, w_n)$.

Definition 12.4. Let $\Phi : V \rightarrow W$ be an isomorphism of vector spaces with orientations Λ_1 and Λ_2 on V and W respectively. Then Φ is said to be *orientation-preserving* if an ordered basis of V in Λ_1 induces an ordered basis of W in Λ_2 . Consider $V = \mathbb{R}^n$. Then the orientation Λ^+ , defined by the natural ordered basis $e_1, \dots, e_n$, where $e_i = (0, \dots, 0, 1, 0, \dots, 0)$ for $i = 1, \dots, n$ is called the *positive orientation* of V .

Definition 12.5. Let X be an n -dimensional manifold and let (U, f) be a chart. The positive orientation Λ^+ on (U, f) is the collection of orientations on $T_x X$ for each $x \in U$, such that, considering the positive orientation on $f(U)$, the isomorphism $\Delta_f : T_x U \rightarrow \mathbb{R}^n$ is orientation preserving. Then X is called *orientable* if there exists an atlas $\{(U_i, f_i)\}$ of positively oriented charts, such that for $x \in U_i \cap U_j$, the orientation induced by (U_i, f_i) coincides with the orientation induced by (U_j, f_j) . An orientable manifold with a choice of an orientation is called *oriented manifold*.

Remark. In such case, all the transition functions have differential with positive determinant.

Definition 12.6. Let $F : X \rightarrow Y$ be a smooth function between oriented manifold. Then F is said to be *orientation preserving* if for any $x \in X$, the linear map $DF|_x : T_x X \rightarrow T_{F(x)} Y$ is orientation preserving, with respect to the positive orientations induced on $T_x X$ and $T_{F(x)} Y$ respectively.

Definition 12.7. Let X be a manifold. For any $p \geq 0$, we denote $\Omega_c^p(X) := \{\omega \in \Omega^p(X) \mid \omega \text{ has compact support}\}$. Assume that $\omega \in \Omega_c^n(X)$ and that $\text{Supp}(\omega) \subset U$ where (U, f) is a positively oriented chart of X , then $(f^{-1})^* \omega \in \Omega_c^n(f(U))$ and

$$\int_X \omega := \int_{f(U)} (f^{-1})^* \omega$$

This integral is well-defined, i.e. it doesn't depend on the chart chosen, since if we have two positively oriented charts $(U, f), (U', f')$ containing $\text{Supp}(\omega)$, then the differential of the transition function $\phi = f' \circ f^{-1} : f(U \cap U') \rightarrow f'(U \cap U')$ has positive determinant. Thus

$$\begin{aligned} \int_{f'(U')} (f'^{-1})^* \omega &= \int_{f'(U \cap U')} (f'^{-1})^* \omega = \int_{f(U \cap U')} (f' \circ f^{-1})^* (f'^{-1})^* \omega \\ &= \int_{f(U \cap U')} (f^{-1})^* f'^* (f'^{-1})^* \omega = \int_{f(U \cap U')} (f^{-1})^* \omega = \int_{f(U)} (f^{-1})^* \omega \end{aligned}$$

Definition 12.8. Let X be a manifold and let $\mathcal{U} = \{U_i\}$ be an open covering. A *partition of unity* with respect to $\mathcal{U}$ is a collection of smooth functions $h_i : X \rightarrow [0, 1]$ such that

- (1) $\text{Supp}(h_i) \subset U_i$ for all i (in particular, $h_i = 0$ outside U_i)
- (2) $\sum_i h_i(x) = 1$ for all $x \in X$
- (3) for all $x \in X$ there exists an open nbhd V of x such that $\text{Supp}(h_i) \cap V \neq \emptyset$ for only finitely many i .

Fact 12.9. If X is a manifold, any open cover of X induces a partition of unity.

Proposition 12.10. Let X be a manifold of dimension n . Then X is orientable if and only if there exists a non-vanishing n -form $\omega \in \Omega^n(X)$. Let ω be a non-vanishing n -form that has the same orientation as X , then ω is called a *volumn form*.

Proof. Assume that ω is a non-vanishing n -form on X and let $\{(U_i, f_i)\}_{i \in I}$ be an atlas on X . For each i , we may write $(f_i^{-1})^*\omega = g_i dx_1 \wedge \cdots \wedge dx_n$ where g_i is a smooth function on $f_i(U_i)$. Since ω is non-vanishing, it follows that g_i is either positive or negative on $f_i(U_i)$. After replacing (if necessary) f_i by $t \circ f_i$ where $t : \mathbb{R}^n \rightarrow \mathbb{R}^n : (x_1, \dots, x_n) \mapsto (-x_1, x_2, \dots, x_n)$, we may assume that g_i is positive for all i . Indeed, this is because $(f_i^{-1} \circ t^{-1})^*\omega = (t^{-1})^* \circ (f_i^{-1})^*\omega = t^*(g_i dx_1 \wedge \cdots \wedge dx_n) = -g_i dx_1 \wedge \cdots \wedge dx_n$. Now, if we consider the orientation on U_i s.t. (U_i, f_i) is positively oriented, then if $x \in U_i \cap U_j$ for some $i, j \in I$, then U_i, U_j define the same orientation on $U_i \cap U_j$ since $(f_j^{-1})^*\omega = g_j dy_1 \wedge \cdots \wedge dy_n = \phi_{ij}^*(f_i^{-1})^*\omega = g_i \circ \phi_{ij} \cdot \det D\phi_{ij} dy_1 \wedge \cdots \wedge dy_n$. This implies that $\det D\phi_{ij}$ is positive and thus X is orientable.

Assume that X is orientable and let $\{(U_i, f_i)\}_{i \in I}$ be an atlas of positively oriented charts. Let $h_i : X \rightarrow [0, 1]$ be a partition of unity with respect to the atlas and define $\omega_i = f_i^*(dx_1 \wedge \cdots \wedge dx_n)$. Let $\tilde{\omega}_i$ be the n -form defined by ω_i inside U_i and equal to zero outside U_i (such $\tilde{\omega}_i$ is not necessarily a smooth n -form). Then we define a smooth n -form $\omega := \sum_i h_i \tilde{\omega}_i$. For every point, there are only finitely many non-zero terms to sum up so this sum is valid. For each $x \in X$ and for each positively oriented basis $v_1, \dots, v_p \in T_x X$, we have that $\omega(x)(v_1, \dots, v_p) > 0$ since for $x \in U_i$, $\omega_i(x)(v_1, \dots, v_p) > 0$ by positive orientivity of $v_1, \dots, v_p$. So ω is a volumn form on X .

Definition 12.11. Let X be an oriented manifold. Let $\omega \in \Omega_c^n(X)$ and let $\{(U_i, f_i)\}_{i \in I}$ be an atlas if X with positively oriented smooth charts. Let $h_i : X \rightarrow [0, 1]$ be a partition of unity with respect to the atlas. Then the *integral* of ω over X is defined by

$$\int_X \omega = \sum_{i \in I} \int_X h_i \omega$$

For each i , we have that the support of $h_i \omega$ is contained in U_i and therefore each term of the sum is well-defined as in definition 12.7:

$$\int_X \omega = \sum_{i \in I} \int_X h_i \omega = \sum_{i \in I} \int_{f_i(U_i)} (f_i^{-1})^* h_i \omega$$

Lemma 12.12. The integral $\int_X \omega$ does not depend on the choice of the atlas $\{(U_i, f_i)\}_{i \in I}$ and the choice of the partition of unity h_i (w.r.t. the atlas).

Proof. We've shown that for an n -form whose support is contained in a smooth chart, the integral does not depend on the chosen chart. Now let $\{(U_i, f_i)\}_{i \in I}$ and $\{(\bar{U}_j, \bar{f}_j)\}_{j \in J}$ be atlases on X (in the fixed smooth structure of X) defined by positively oriented charts and let $h_i : X \rightarrow [0, 1]$ and $\bar{h}_j : X \rightarrow [0, 1]$ be partition of unity with respect to these coverings. Then, since by definition, $\sum_{i \in I} h_i = \sum_{j \in J} \bar{h}_j = 1$, we have

$$\int_X h_i \omega = \int_X \left(\sum_{j \in J} \bar{h}_j \right) h_i \omega = \sum_{j \in J} \int_X \bar{h}_j h_i \omega$$

(notice here each summand in the last expression is an integral over U_i) and hence

$$\sum_i \int_X h_i \omega = \sum_{i,j} \int_X \bar{h}_j h_i \omega = \sum_j \int_X \bar{h}_j \left(\sum_i h_i \right) \omega = \sum_j \int_X \bar{h}_j \omega$$

Here, the summands in the last expression are indeed integrals over $\bar{U}_j$, and the equality is due to remarks after 12.7. (the positive orientation of the manifold is already fixed) Thus, the integral $\int_X \omega = \sum_i \int_X h_i \omega$ does not depend on the choice of charts the partition of unity.

Proposition 12.13. Let X and Y be oriented manifolds of dimension n and let $\omega, \eta \in \Omega_c^n(X)$. Then the following properties hold:

(Linearity) $\int_X (a\omega + b\eta) = a \int_X \omega + b \int_X \eta$ for all $a, b \in \mathbb{R}$.

(Orientation reversal) Let $\bar{X}$ denote the manifold X with opposite orientation. Then $\int_{\bar{X}} \omega = -\int_X \omega$

(Positivity) Let ω be a volume form on X , then $\int_X \omega > 0$.

(Diffeomorphism invariance) Let $F : Y \rightarrow X$ be an orientation preserving diffeomorphism. Then

$$\int_X \omega = \int_Y F^* \omega$$

Proof. (1) The first property is a direct consequence from the definition 12.11.

(2) For orientation reversal, the determinants of the Jacobians of the transition functions between two charts on two atlases (defined by charts in opposite orientations) would be negative.

(3) Let ω be a volume form. Then for each atlas $\{(U_i, f_i)\}$ and for each partition of unity h_i , the form $(f_i^{-1})^* \omega$ is locally of the form $g dx_1 \wedge \cdots \wedge dx_n$ for some local coordinate $(x_1, \dots, x_n)$, where g is a smooth function such that $g > 0$. Hence each term in $\sum_i \int_{f_i(U_i)} (f_i^{-1})^* h_i \omega = \sum_i \int_{f_i(U_i)} h_i g dx_1 \wedge \cdots \wedge dx_n$ is non-negative and at least one term is strictly positive. Thus $\int_X \omega > 0$.

(4) For diffeomorphism invariance, let $F : Y \rightarrow X$ be an orientation preserving diffeomorphism. Take an atlas on X defined by positively oriented charts $\{(U_i, f_i)\}_{i \in I}$, then a positively oriented atlas for Y is $\{(F^{-1}(U_i), f_i \circ F)\}_{i \in I}$. If h_i is the partition of unity on X , then we may take $h_i \circ F$ the partition of unity on Y . Now $\int_X \omega = \sum_{i \in I} \int_{f_i(U_i)} (f_i^{-1})^* h_i \omega$ by definition, and

$$\begin{aligned} \int_Y F^* \omega &= \sum_{i \in I} \int_{f_i(U_i)} (F^{-1} \circ f_i^{-1})^* h_i \circ F \cdot F^* \omega = \sum_{i \in I} \int_{f_i(U_i)} h_i \circ F \circ F^{-1} \circ f_i^{-1} \cdot (F^{-1} \circ f_i^{-1})^* F^* \omega \\ &= \sum_{i \in I} \int_{f_i(U_i)} h_i \circ f_i^{-1} \cdot (f_i^{-1})^* \omega = \int_X \omega \end{aligned}$$

Definition 12.14. We consider the set $\mathbb{R}_+^n := \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_n \geq 0\}$ with the topology induced by $\mathbb{R}^n$. The boundary of $\mathbb{R}_+^n$ is the set $\partial \mathbb{R}_+^n := \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_n = 0\}$. Define a *manifold with boundary of dimension n* is a Hausdorff and second countable topological space X with an atlas $\{(U_i, f_i)\}$, such that the functions $f_i : U_i \rightarrow \mathbb{R}_+^n$ is a homeomorphism for any i and, for each i and j , the induced morphism

$$f_i \circ f_j^{-1} : f_j(U_i \cap U_j) \rightarrow f_i(U_i \cap U_j)$$

is a smooth morphism. The *boundary* ∂X of X is defined as

$$\partial X := \{x \in X \mid f_i(x) \in \partial \mathbb{R}_+^n\}$$

The *interior* of X is the set $\text{int}(X) := X \setminus \partial X$

Remarks 12.15

- The definition of ∂X does not depend on the chart chosen. (This is a result in algebraic topology)
- If X is a manifold with boundary then the boundary of X is a closed subset of X . In particular if X is compact then ∂X is compact.
- if X is a manifold with boundary of dimension n then $\text{int}(X)$ is a manifold of dimension n .
- Manifolds are defined intrinsically, meaning that they are not defined as subsets of another topological space, therefore, the notion of boundary will differ from the usual boundary of a topological subset.
- The notions of the tangent space, vector fields, orientability, differential forms, partition of unity,... are defined for manifolds with boundary in exactly the same way as for ordinary manifolds.
- If X is an oriented manifold then ∂X is also oriented. Recall that if $X = \mathbb{R}^n$, the volume form $dx_1 \wedge \cdots \wedge dx_n$ defines a positive orientation on X . Similarly, for each point in the interior of $\mathbb{R}_+^n$, the volume form $dx_1 \wedge \cdots \wedge dx_n$ defines a positive orientation and this (as a convention) induces the positive volume form on ∂X given by $(-1)^n dx_1 \wedge \cdots \wedge dx_{n-1}$.

Examples 12.16.

- $X = [0, 1] \subset \mathbb{R}$ is a manifold with boundary and $\partial X = \{0, 1\}$.

- The closed disc $D = \{x \in \mathbb{R}^n \mid |x| \leq 1\}$ is a manifold with boundary $\partial D = \{x \in \mathbb{R}^n \mid |x| = 1\} = S^{n-1}$.
- The cylinder defined as the product of a real interval with the circle $X = [0, 1] \times S^1$ is a manifold with boundary and ∂X is the union of two circles, i.e. a manifold with 2 components.
- It's not true that the product of 2 manifolds with boundary is still a manifold with boundary since $[0, 1] \times [0, 1]$ has corner.

Fact 12.17. Let X be a manifold with boundary ∂X and let $\mathcal{U} = \{U_i\}$ be an open cover of X . Any manifold with boundary X , with open cover $\mathcal{U} = \{U_i\}$ admits a partition of unity with respect to $\mathcal{U}$. Similarly, if X is a manifold with boundary of dimension n and $\omega \in \Omega_c^n(X)$ then the integral $\int_X \omega$ of ω is defined.

Theorem 12.18. (Stokes' theorem) Let X be a smooth oriented manifold with boundary and let $\omega \in \Omega_c^{n-1}(X)$, then

$$\int_X d\omega = \int_{\partial X} \omega$$

Proof. Let $\{(U_i, f_i)\}_{i \in I}$ be an atlas on X and let $h_i : X \rightarrow [0, 1]$ be a partition of unity with respect to this covering. Then

$$\int_X d\omega = \sum_{i \in I} \int_X d(h_i \omega) = \sum_{i \in I} \int_{f_i(U_i)} (f_i^{-1})^* d(h_i \omega)$$

But we know that taking pull-back and taking de Rham differential commutes with each other, so $(f_i^{-1})^* d(h_i \omega) = d(f_i^{-1})^* h_i \omega$. Let $\tilde{h}_i = h_i \circ f_i^{-1}$, then the $(n-1)$ -form $(f_i^{-1})^* h_i \omega$ in $\mathbb{R}_+^n$ can be written in local coordinates on $f_i(U_i)$ as $\sum_{j=1}^n \tilde{h}_i \omega_j dx_1 \wedge \cdots \wedge \widehat{dx_j} \wedge \cdots \wedge dx_n$ where $\omega_j \in C^\infty(f_i(U_i))$ and $\widehat{dx_j}$ means that we omit the component dx_j . The differential of this form is then given by

$$\begin{aligned} d \left(\sum_{j=1}^n \tilde{h}_i \omega_j dx_1 \wedge \cdots \wedge \widehat{dx_j} \wedge \cdots \wedge dx_n \right) &= \sum_{j=1}^n \frac{\partial (\tilde{h}_i \omega_j)}{\partial x_j} dx_j \wedge dx_1 \wedge \cdots \wedge \widehat{dx_j} \wedge \cdots \wedge dx_n \\ &= \left(\sum_{j=1}^n (-1)^{j-1} \frac{\partial (\tilde{h}_i \omega_j)}{\partial x_j} \right) dx_1 \wedge \cdots \wedge dx_n \end{aligned}$$

Thus, we have

$$\int_X d\omega = \sum_i \underbrace{\int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty}}_{n-1 \text{ times}} \int_0^\infty \sum_{j=1}^n (-1)^{j-1} \frac{\partial (\tilde{h}_i \omega_j)}{\partial x_j} dx_n \cdots dx_1$$

where the i -th integral of the first $(n-1)$ integrals correspond to the integration over dx_i , for $i = 1, \dots, n-1$, and the last integral corresponds to the integration over dx_n . Now we may apply Fubini's theorem and integrate over dx_j first. Thus by the fundamental theorem of calculus, we have

$$\int_X d\omega = \sum_i \sum_{j=1}^n \int_{-\infty}^{+\infty} \cdots \widehat{\int_{-\infty}^{+\infty}} \cdots \int_{-\infty}^{+\infty} \int_0^\infty (-1)^{j-1} (\tilde{h}_i \omega_j) \Big|_{\text{at boundary}} dx_n \cdots \widehat{dx_j} \cdots dx_1$$

where we have to evaluate the result of each integral at the boundary of the domain of integration. But on this boundary if U_i is a chart of the interior of X , then h_i has to be zero and on the boundary charts, only the part of the boundary of the chart which touches the boundary of X (i.e. where $x_n = 0$) will have $h_i \neq 0$. The extra minus sign comes from the fact that the non-vanishing term of the integral is its lower bound.

Thus by convention we have

$$\int_X d\omega = \sum_i \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} (-1)^n \tilde{h}_i(x_1, \dots, x_{n-1}, 0) \omega_n dx_1 \wedge \cdots \wedge dx_{n-1} = \int_{\partial X} \omega$$

The last equality follows from the following commutative diagram: $\int_{\partial X} i^* \omega = \sum_i \int_{g_i(U_i \cap \partial X)} (g_i^{-1})^* h_i i^* \omega$

$$\begin{aligned} &= \sum_i \int_{g_i(U_i \cap \partial X)} \tilde{h}_i(x_1, \dots, x_{n-1}, 0) (i \circ g_i^{-1})^* \omega = \sum_i \int_{g_i(U_i \cap \partial X)} \tilde{h}_i(x_1, \dots, x_{n-1}, 0) (f_i^{-1} \circ \tilde{i})^* \omega \\ &= \sum_i \int_{g_i(U_i \cap \partial X)} \tilde{h}_i(x_1, \dots, x_{n-1}, 0) \tilde{i}^* \sum_{j=1}^n \omega_j dx_1 \wedge \cdots \wedge \widehat{dx_j} \wedge \cdots \wedge dx_n \end{aligned}$$

$$= \sum_i \int_{g_i(U_i \cap \partial X)} \tilde{h}_i(x_1, \dots, x_{n-1}, 0) \sum_{j=1}^n \omega_j(x_1, \dots, x_{n-1}, 0) \tilde{i}^* dx_1 \wedge \dots \wedge \widehat{dx_j} \wedge \dots \wedge dx_n$$

and the only non-vanishing term is $\sum_i \int_{g_i(U_i \cap \partial X)} \tilde{h}_i(x_1, \dots, x_{n-1}, 0) \omega_n(x_1, \dots, x_{n-1}, 0) dx_1 \wedge \dots \wedge dx_{n-1}$ so the statement is true, $(-1)^n$ comes from our convention.

$$\begin{array}{ccc} U_i & \xrightarrow{f_i} & f_i(U_i) \\ i \uparrow & \circ & \uparrow \tilde{i} \\ U_i \cap \partial X & \xrightarrow{g_i} & f_i(U_i) \cap \partial \mathbb{R}_+^n \end{array}$$

Corollary 12.19. (Integration by parts) Let X be a manifold with boundary of dimension n and $p \geq 0$. Let $\omega \in \Omega_c^p(X)$ and $\eta \in \Omega_c^{n-p-1}(X)$. Then $\int_{\partial X} \omega \wedge \eta = \int_X d\omega \wedge \eta + (-1)^p \int_X \omega \wedge d\eta$.

Proof. By Stoke's theorem and Leibniz rule, $\int_{\partial X} \omega \wedge \eta = \int_X d(\omega \wedge \eta) = \int_X d\omega \wedge \eta + (-1)^p \int_X \omega \wedge d\eta$.

Corollary 12.18. (Brouwer's fixed point theorem) Let $D := \{x \in \mathbb{R}^n \mid |x| \leq 1\}$ be the closed disc and let $f : D \rightarrow D$ be a smooth morphism. Then there exists $x \in D$ such that $f(x) = x$.

Proof. Suppose by contradiction that $f(x) \neq x$ for all $x \in D$. Then the ray beginning at $f(x)$ and passing through x is uniquely defined. Let $g(x)$ be the point that the ray meets at the boundary of D . Then $g : D \rightarrow \partial D$ is a smooth function such that $g(x) = x$ for all $x \in \partial D$. Note that ∂D is the $(n-1)$ -dimensional sphere, which is orientable. Thus, it admits a volume form $\omega \in \Omega^{n-1}(\partial D)$. Thus,

$$0 < \int_{\partial D} \omega = \int_{\partial D} g^* \omega = \int_D d(g^* \omega) = \int_D g^* d\omega$$

where the first equality comes from the fact that g is the identity on ∂D , the second from Stokes' theorem and the third follows by the property of the de Rham differential. But $d\omega$ is a n -form on the $(n-1)$ -dimensional manifold ∂D so it is zero. Thus, we get a contradiction.

Lemma 12.19. A closed form on a manifold X is not necessarily exact.

Proof. Let $X = \mathbb{R}^2 \setminus \{(0,0)\}$ and let $\omega = \frac{x}{x^2+y^2} dy - \frac{y}{x^2+y^2} dx$ be a one-form on X . Then $d\omega = \frac{\partial}{\partial x}(\frac{x}{x^2+y^2}) dx \wedge dy - \frac{\partial}{\partial y}(\frac{y}{x^2+y^2}) dy \wedge dx = 0$. Thus ω is closed. Let $\gamma : [0, 2\pi] \rightarrow X : \theta \mapsto (\cos \theta, \sin \theta) \in X$. Then

$$\int_{S^1} \omega = \int_{[0, 2\pi]} \gamma^* \omega = \int_0^{2\pi} \frac{\cos^2 \theta - \sin \theta (-\sin \theta)}{\cos^2 \theta + \sin^2 \theta} d\theta = 2\pi$$

Assume by contradiction that ω is exact. Then there exists a 0-form, i.e. a smooth function f on X such that $\omega = df$. But S^1 has no boundary, so by Stokes' theorem, $\int_{S^1} \omega = \int_{S^1} df = \int_{\partial S^1} f = 0$.

Proposition 12.20. Let $\omega \in \Omega_c^n(X)$ be an exact form on an oriented manifold X of dimension n without boundary. Then $\int_X \omega = 0$.

Proof. $\int_X \omega = \int_X df = \int_{\partial X} f = 0$.

Proposition 12.21. Let X be an oriented manifold of dimension n with boundary and let $\omega \in \Omega_c^{n-1}(X)$ be a closed form. Then $\int_{\partial X} \omega = 0$.

Proof. $\int_{\partial X} \omega = \int_X d\omega = 0$.

Definition 12.22. Let X be an oriented manifold of dimension n and let $Z \subset X$ be a oriented submanifold of dimension k . Let $\omega \in \Omega_c^k(X)$. Then we define the integral of ω on Z as

$$\int_Z \omega := \int_Z i^* \omega$$

where $i : Z \rightarrow X$ is the inclusion map. Note that $i^* \omega \in \Omega^k(Z)$. We denote

$$\omega|_Z := i^* \omega$$

Proposition 12.23. Let X be an oriented manifold of dimension n and let $\omega \in \Omega_c^k(X)$. Let $Z \subset X$ be a compact oriented submanifold of dimension k without boundary and suppose $\int_Z \omega \neq 0$. Then

- (1) ω is not exact on X and $\omega|_Z$ is not exact on Z .
- (2) Z is not the boundary of a compact oriented submanifold $Y \subset X$ of dimension $k+1$.

Proof. See coursework 5.